NOVVES

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SOLUTION 1
Smoke and Heat Exhaust Systems

Enclosed Car Parks

Road and Railway Tunnels

Shopping Malls

Stadiums and Assembly Halls

Industrial Manufacturing Facilities

Hospitals and Healthcare Facilities

Airports

Warehousing and Logistics Centers

Power Plants & Generator Rooms

Hotels and Residential High-Rise Buildings

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“born to *flow*.”

NOVVES is more than a manufacturer of air movement solutions; it is a system architect that restores airflow, safeguards continuity, and enables spaces to breathe even in the most demanding and uncertain conditions.

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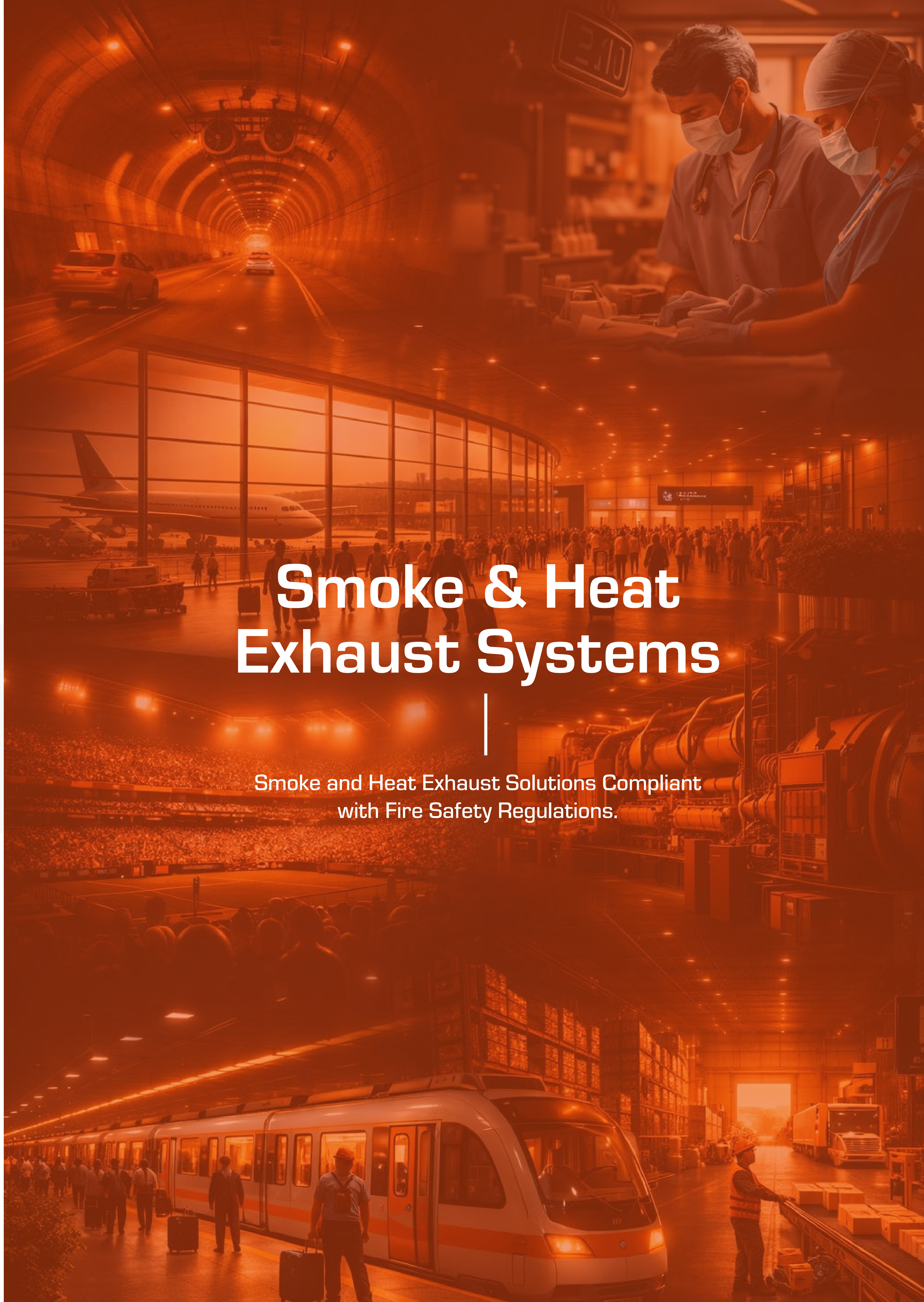
Smoke & Heat Exhaust Systems
Smoke and Heat Exhaust Solutions Compliant with Fire Safety Regulations.

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UNDERGROUND CAR PARKS

Dragonfly: Jet Fans, Axial Smoke & Heat Exhaust Fans
Marlin: Axial Fresh Air Supply Fans
Hound: Smoke Control Dampers
Hawk: Automation & Control Panels

1.1

UNDERGROUND CAR PARKS

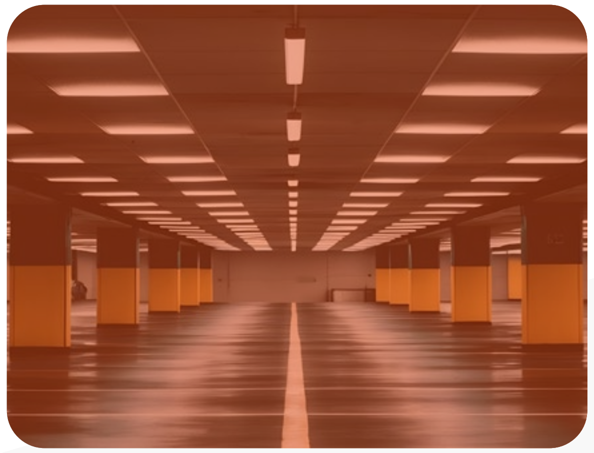
Smoke and heat exhaust systems for underground car parks are designed to provide mechanical ventilation during normal operation, maintaining vehicle-emission pollutants such as CO and NOx within safe limits. In fire scenarios, the system controls and extracts smoke and heat, protecting escape routes from smoke contamination and enabling safer, more effective firefighting intervention.

Engineering Principles, Technological Components and Safety Standards of Smoke and Heat Exhaust Systems for Underground Car Parks

Modern urban development, limited land availability, and the continuous growth in vehicle ownership have made underground parking facilities an essential part of contemporary infrastructure. Due to their architectural characteristics, underground car parks are complex engineering environments with limited natural ventilation, restricted air circulation, and a high fire-risk potential. Within these facilities, smoke and heat exhaust systems are far more than mechanical installations; they are critical life-safety systems positioned at the core of a building's passive and active fire protection strategy. During normal operation, these systems dilute and remove toxic vehicle-emission pollutants, while in the event of a fire, they control and extract smoke to maintain tenable conditions, support safe evacuation, and facilitate effective firefighting operations.

Ventilation Requirements and Engineering Fundamentals for Underground Car Parks

The ventilation of enclosed and underground car parks is based on two fundamental functional requirements. The first is to ensure that pollutants generated by the operation of internal combustion engines—particularly carbon monoxide (CO) and nitrogen oxides (NOx)—are maintained at levels that do not pose a risk to human health during normal operation. The second, and most critical, requirement is the controlled management of high-temperature smoke and toxic gases generated during a fire, creating conditions that enable safe occupant evacuation and allow firefighting teams to carry out effective emergency response operations.



1. Daily Ventilation and Indoor Air Quality Management

Internal combustion engines emit carbon monoxide (CO), nitrogen oxides (NOx), and various hydrocarbon compounds during operation. Among these pollutants, carbon monoxide is considered the most hazardous due to its affinity for hemoglobin, which is approximately 200–250 times greater than that of oxygen. By binding to hemoglobin, CO reduces the blood's oxygen-carrying capacity and prevents adequate oxygen delivery to body tissues, resulting in hypoxia. In enclosed environments, elevated CO concentrations can lead to symptoms ranging from headaches, dizziness, and loss of consciousness to potentially fatal oxygen deprivation at higher exposure levels. For this reason, daily ventilation systems are designed to continuously monitor and maintain CO concentrations below predefined threshold values, typically **between 35 and 50 ppm**. To maximize energy efficiency, these systems operate through sensor-based control strategies. CO and **NOx** sensors installed throughout the car park continuously measure air quality and transmit real-time data to the central automation and control panel **(NOVVES Hawk).** Based on the detected pollutant levels, the automation system adjusts fan operation across multiple speed stages or activates jet fans only within the affected zones where contamination levels increase. This demand-controlled ventilation approach prevents the system from operating continuously at full capacity, reducing energy consumption while minimizing mechanical wear and extending equipment service life.

Engineering in underground car parks is driven by safety, not comfort.

2. Smoke and Heat Management During Fire Emergencies

Underground car park fires are characterized by the generation of large quantities of smoke and heat due to the presence of plastics, synthetic materials, and batteries used in modern hybrid and electric vehicles. During a fire, smoke rapidly mixes with air, spreads throughout the space, and can reduce visibility to near zero within minutes. The primary objective of a smoke exhaust system is to prevent smoke logging throughout the entire car park volume by creating a controlled smoke layer beneath the ceiling and extracting this layer through designated exhaust shafts. The effectiveness of the system is evaluated based on tenability criteria, which define the conditions required to support safe evacuation and emergency response. Among these criteria, visibility is one of the most critical parameters. Maintaining adequate visibility allows occupants to identify escape signage, navigate toward exits without panic, and evacuate safely. Furthermore, controlling smoke temperatures helps protect the building's reinforced concrete or steel structural elements from heat-induced damage, reducing the risks of structural degradation, spalling, and potential collapse.

Regulatory Requirements in Türkiye

Regulation on Fire Protection of Buildings (BYKHY):

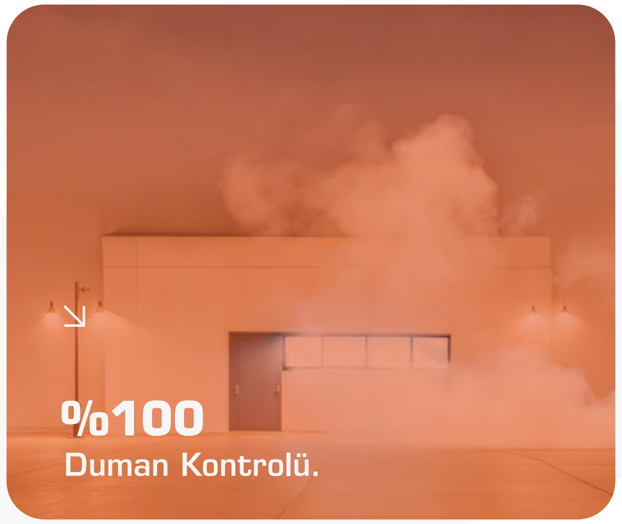
For underground car parks with a total area exceeding 2,000 m², a mechanical smoke exhaust system is mandatory.
A minimum of 10 air changes per hour (ACH) must be provided.
The system must operate independently from other building ventilation systems.
Equipment must be supplied through fire-resistant cabling.
Power supply continuity must be maintained under emergency conditions.

Note: The 2,000 m² threshold is a national requirement specific to Türkiye. Requirements may vary across European countries. For example, in Germany, thresholds may differ between federal states, while in the United Kingdom, smoke control systems are generally designed using a risk-based approach. In many U.S. applications, underground car parks are required to be equipped with mechanical ventilation systems.

How Does the System Work?

Jet fans generate high-velocity air jets that drive smoke toward designated exhaust shafts. Through the induction effect, they entrain surrounding air and move airflow volumes several times greater than the fan's own capacity. As a result:

- Smoke spread (smoke logging) is controlled.
- Visibility on escape routes is maintained.
- Safe evacuation conditions are preserved.
- Firefighting access remains clear.
- Structural integrity is protected.
- Back-layering is prevented.



Comparative Analysis of Jet Fan and Ducted Systems

Traditional ducted ventilation systems are increasingly being replaced by impulse ventilation systems based on jet fan technology. This transition provides significant structural, operational, and energy-efficiency advantages.

Structural & Architectural Benefits:
Ducted systems require extensive ductwork throughout the car park, reducing available ceiling height. Jet fan systems minimize ductwork requirements, allowing more efficient use of space and reducing construction costs.

Energy & Operational Efficiency:
In ducted systems, fans must overcome high duct resistance, requiring higher static pressures. Jet fan systems operate at significantly lower pressures, reducing installed power requirements and energy consumption. During daily operation, ventilation can also be activated only in affected zones, further improving efficiency.

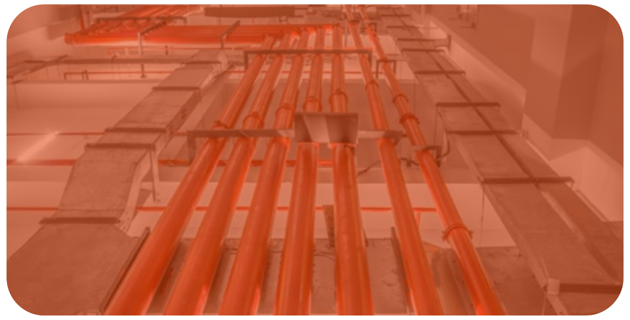
CRITERION	DUCT SYSTEM	JET FAN SYSTEM
Installation Time	Time-consuming / Long-term installation	Fast Installation
Fire Safety / Smoke Evacuation Risk	High risk of uncontrolled smoke spread	Directs smoke to evacuation shafts
Aesthetics and Visibility	Bulky and complex	Clean and unobtrusive
Ceiling Height	Reduced due to ductwork	20-30 cm higher ceiling
Static Pressure	600-1000 Pa (high)	200-300 Pa (low)
Installed Motor Power	High	3-4 floors lower
Transformer - Generator Capacity	Large	Much smaller
Air Quality Distribution	Limited effective area, dead zones exist	Homogeneous distribution, no dead zones
Construction Cost	High (more excavation and concrete)	Lower
Energy Consumption (Daily)	The entire system runs	Only dirty zones (60-70% savings)
Maintenance	Duct cleaning is difficult	Jet fans with periodic control

Coordination with Sprinkler Systems

- Smoke control systems are designed to operate in coordination with sprinkler systems:
- Sprinklers suppress and limit fire growth but do not control smoke.
- By reducing the fire load, sprinkler systems decrease the rate of smoke production.
- Under sprinkler-protected conditions, a design temperature limit of 60°C can be applied (115°C for non-sprinklered scenarios).
- NFPA 88A (2023): Automatic sprinkler protection is now required in all parking structures, including open parking garages.

System Components and NOVVES Product Technologies

A high-performance smoke and heat exhaust system consists of four integrated layers working in complete coordination: the impulse (momentum) layer, exhaust layer, make-up air layer, and control (automation) layer.



System Components and NOVVES Product Technologies

Dragonfly: Axial and Centrifugal Fan Technology
The Dragonfly series includes fans designed to perform both daily ventilation and emergency smoke extraction duties. Certified in accordance with EN 12101-3, these fans are available in F300 and F400 classifications, ensuring continuous operation for up to 2 hours at 300°C or 400°C under demanding fire conditions.

1. Axial Jet Fans (JAU/JAR): Axial jet fans generate airflow momentum and direct air masses toward exhaust shafts. Models are available in both single-direction (JAU) and reversible (JAR) configurations. Reversible units can fully reverse airflow direction at 100% capacity according to the fire scenario, providing maximum design flexibility.

2. Centrifugal Jet Fans (JR): Centrifugal jet fans are particularly suitable for car parks with limited ceiling heights. Their slim and compact construction minimizes installation space requirements while maintaining vehicle clearance..

3. Smoke Exhaust Fans (Dragonfly T/R/C): These primary exhaust fans, installed on shafts or rooftops, discharge smoke transported by the jet fan system to the outside environment. Available as Tube Axial (T), Roof Mounted (R), and Cabinet Type (C) versions, they provide project-specific installation flexibility.

Hound: Smoke Control Dampers and Airflow Management

One of the most critical elements of a smoke control system is the damper, which prevents smoke from entering unintended shafts or migrating into protected areas. The Hound series is engineered for high airtightness and reliable fire resistance performance.

• **Hound SD M24 (Shaft Smoke Damper):** Tested and certified in accordance with EN 12101-8, the Hound SD M24 is designed for use in single-compartment smoke control systems. Its EN 1751 Class C airtightness rating prevents smoke leakage into unintended shafts and ensures that smoke is extracted only through designated exhaust routes.

• **Materials and Safety:** The damper incorporates high-temperature-resistant materials such as calcium silicate. Intumescent seals positioned along the blade edges expand when exposed to fire, providing enhanced smoke containment and airtight sealing under emergency conditions.

Marlin and Hawk: Make-Up Air Supply and Intelligent Control

System continuity depends on the controlled supply of fresh air to replace the extracted smoke. Marlin axial fans deliver fresh air into the car park through designated supply points such as ramps or shafts, creating a balanced airflow pattern throughout the facility. The integration of all mechanical equipment is managed by the Hawk control system. Hawk processes signals received from the fire detection system and activates a predefined Fire Scenario Matrix. Based on the fire location and zone, the system determines within seconds which jet fans should operate, which dampers should open, and at what duty the main exhaust fans should run.

Equipmant Group	Product Series	Functional Role	Key Technical Feature
Jet Fans	Dragonfly JAR/JAU/JR	Smoke guidance and impulse generation	F300 -F400 (300-400°C/120 min) Certified
Smoke & Heat Exhaust Fans	Dragonfly T/R/C	Extracting smoke and heat from the building	EN 12101-3 Compliant
Smoke Dampers	Hound SD	Shaft control and smoke-tight sealing	Class-C Leakage Rating / EN 12101-8
Fresh Air Fans	Marlin	Make-up air supply and pressure balancing	Compact tube-axial design / EN 1366-10
Control Panels	Hawk	Scenario management and BMS integration	CO/NOx and fire detection interface

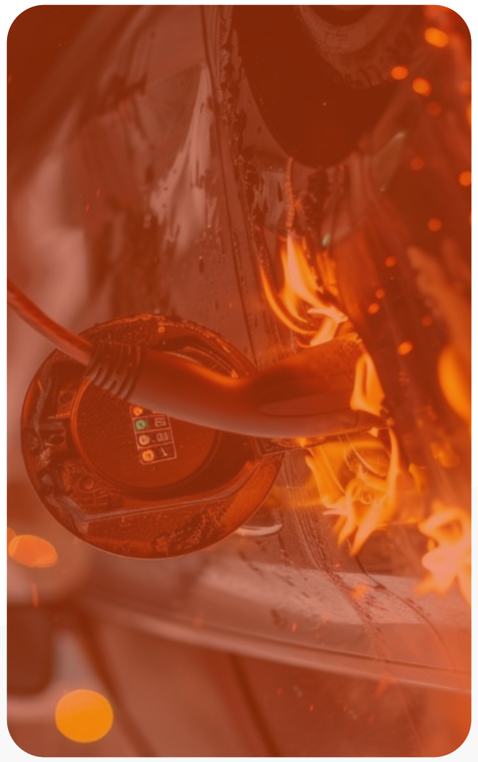
Fire Scenarios and Critical Risk Analysis

Underground car park fires can behave differently depending on the point of ignition, vehicle type, and architectural characteristics of the facility. Therefore, the ventilation design must be capable of addressing a wide range of fire scenarios.

Localized and Developing Fires:

A fire originating in a single vehicle (localized fire) can often be controlled by rapidly capturing the smoke and directing it toward the nearest exhaust shaft. However, if sprinkler performance is insufficient or delayed, the fire may spread to adjacent vehicles. In such cases, the smoke production rate increases significantly, requiring the smoke control system to operate at full capacity. As smoke temperatures rise, the stack effect becomes more pronounced, increasing the tendency of smoke to migrate through ramps and vertical openings toward upper levels or adjacent areas. Jet fan systems are specifically designed to counteract this phenomenon by creating controlled airflow barriers that physically prevent smoke from spreading beyond the designated smoke control zone.

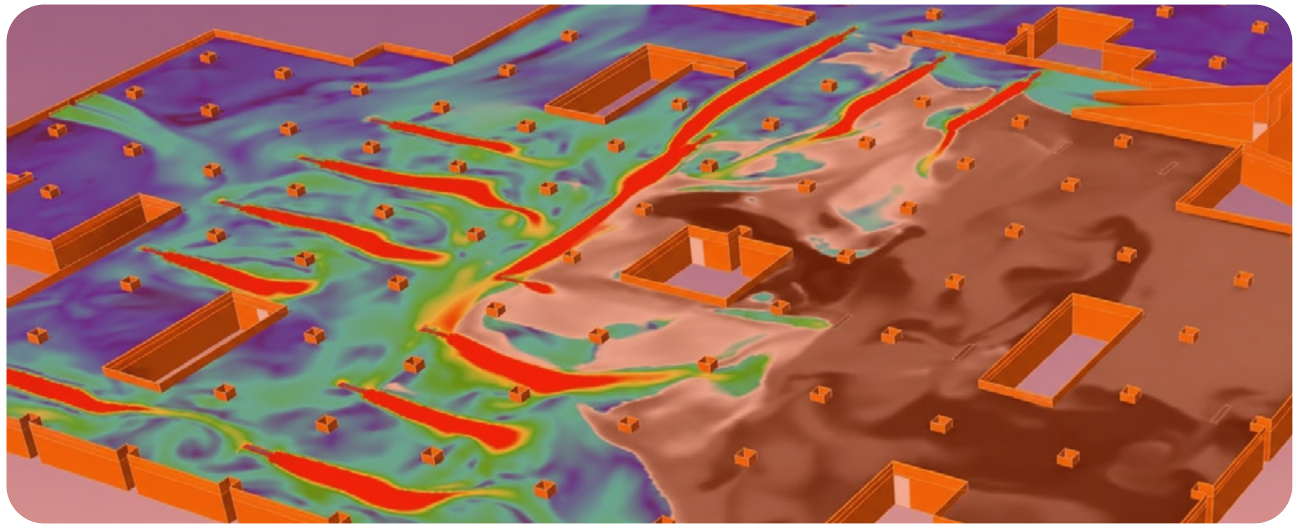
Electric Vehicle (EV) and Hybrid Vehicle Fire:

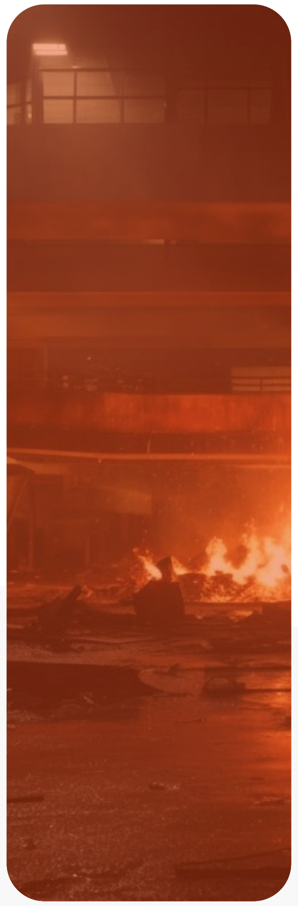


One of the most demanding challenges in modern ventilation design is electric vehicle fires. When lithium-ion batteries ignite, they may enter a process known as thermal runaway. During this event, battery cells generate their own oxygen, making the fire extremely difficult to suppress using conventional firefighting methods and allowing it to continue for extended periods, sometimes for days. EV fires can release highly toxic and corrosive gases such as hydrogen fluoride (HF) and phosphorus pentafluoride (PF_5). These emissions increase both the density and hazard level of the smoke. As a result, current design practices often require increased smoke extraction capacity and a higher concentration of jet fans in areas where EV charging stations are located. Furthermore, because toxic gases may be released before visible smoke is produced, the integration of gas detection sensors into fire detection systems has become a critical safety requirement.

The Core of Design: CFD (Computational Fluid Dynamics) Analysis

Underground car parks are highly complex environments where airflow behavior is difficult to predict due to structural columns, deep beams, and intricate ramp configurations. For this reason, CFD analysis is not a luxury but a necessity in the design and validation of smoke control systems.





Simulation Criteria and Tenability Limits

In CFD analysis, a 3D model of the car park is created and a predefined design fire scenario (e.g., 4 MW or 8 MW) is simulated. The analysis is used to verify the following performance criteria:

- **Smoke Spread and Back-Layering Control:** Verification that the jet fans generate sufficient thrust to prevent smoke from reversing direction due to thermal effects and moving against the airflow.
- **Visibility Analysis:** Assessment of whether visibility along escape routes and evacuation paths remains above 10 meters (or 20 meters under certain standards).
- **Temperature and Thermal Radiation Distribution:** Verification that occupant exposure temperatures remain below 115°C (short-term exposure) or 60°C (long-term exposure / sprinkler-protected environments). In addition, thermal radiation levels must remain below 2.5–3.0 kW/m² to allow firefighting personnel to approach within 10 meters of the fire source.
- **Air Velocity Limitations:** Air velocity at escape doors should not exceed 5 m/s. Higher velocities may hinder occupant movement and interfere with door operation.

Design Criteria

CO Concentration	Max 35 ppm (8-hour average), 100 ppm (20-minute upper limit)
Minimum Visibility Distance	10-20 meters (in escape routes)
Maximum Air Velocity (Door/Ramp)	5 m/s (must not impede human movement)
Maximum Temperature (at 1.8m height)	60°C (sprinklered) / 115°C (short duration)
Thermal Radiation (for evacuation)	< 3 kW/m² (at 10m distance)
Clear Height	Min. 1.8 m (must prevent smoke stratification)
Design Fire Size	4-8 MW (depending on vehicle type)

CFD Analysis Validation Criteria

- Smoke spread rate and direction (smoke logging control)
- Jet fan placement and thrust performance verification
- Back-layering prevention analysis
- Visibility compliance in critical areas (>10 m)
- Temperature distribution and thermal radiation assessment
- Elimination of short-circuit airflow paths
- Identification and mitigation of dead zones
- Stack effect analysis and ramp smoke control

Every CFD report documents compliance with tenability criteria and provides the technical evidence required for authority approval and fire safety verification.

Jet Fan Layout and Exhaust Shaft Strategy

CFD simulations reveal potential short-circuit airflow paths caused by improper jet fan placement. A short circuit occurs when fresh air travels directly to the exhaust shaft without effectively sweeping the car park volume. Through CFD analysis, jet fans are positioned to ensure airflow reaches all critical areas of the car park and directs smoke toward exhaust shafts through the most efficient flow path, maximizing smoke extraction performance and system effectiveness.

Make-Up Air Supply and Pressure Balance

The performance of a smoke exhaust system depends on maintaining a proper balance between the air extracted from the building and the fresh air supplied into it. In underground car parks, a powerful exhaust fan without adequate make-up air can create excessive negative pressure. This may prevent doors—particularly those opening into stairwells and elevator lobbies—from opening properly or cause smoke to migrate into escape routes through leakage paths rather than toward designated exhaust shafts. In an optimized system, the fresh air inlet velocity should be maintained below 2 m/s. This helps prevent turbulence from disrupting the smoke layer and forcing smoke downward. Fresh air is typically supplied through ramps, natural openings, or dedicated Marlin make-up air fans connected to supply shafts. As a general design guideline, the supplied fresh air volume should be approximately 80–90% of the extracted exhaust airflow.

International Standards and Regulatory Compliance

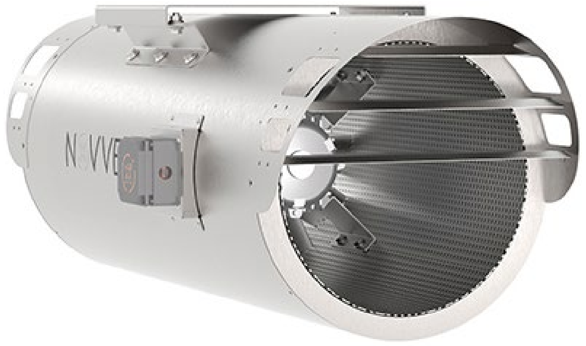
Smoke control systems are governed by stringent regulations worldwide. **NOVVES** solutions are tested and engineered to comply with the highest requirements of international standards and regulations.

Regulation on Fire Protection of Buildings (Türkiye)

According to the current Turkish regulation, underground car parks with a total area exceeding 2,000 m² must be equipped with a mechanical ventilation system. The system must operate independently from other building services and provide a minimum of 10 air changes per hour (ACH). In addition, emergency power supply continuity must be maintained during fire conditions, and all critical equipment must be connected using fire-resistant cabling.

European (EN 12101) and American (NFPA) Standards

The EN 12101 series of European standards defines the performance requirements and testing methodologies for smoke and heat control systems. In particular, EN 12101-3 specifies the high-temperature classifications of smoke exhaust fans, including F200, F300, and F400. In the United States, NFPA 88A focuses on the fire safety requirements of parking structures. The 2023 edition introduced one of the most significant updates by requiring automatic sprinkler protection in all parking facilities, including open parking garages. NFPA 92 remains the primary reference document for the design and calculation of smoke control systems.



Commissioning, Performance Verification and Maintenance

The installation of a smoke exhaust system is not considered complete once construction is finished. Comprehensive testing and verification procedures must be carried out to ensure reliable operation under real fire conditions.

Scenario Testing and Functional Verification

During commissioning, the following checks are performed:

- **Fan Rotation Verification:** Confirmation that each fan operates at the correct speed and in the correct direction according to the fire scenario.
- **Damper Activation Testing:** Verification that the relevant zone dampers move to their designated positions within 60 seconds after a fire signal is received by the control panel.
- **Air Velocity Measurements:** Measurement of airflow velocities at fresh air inlets and exhaust points to verify compliance with CFD design predictions.

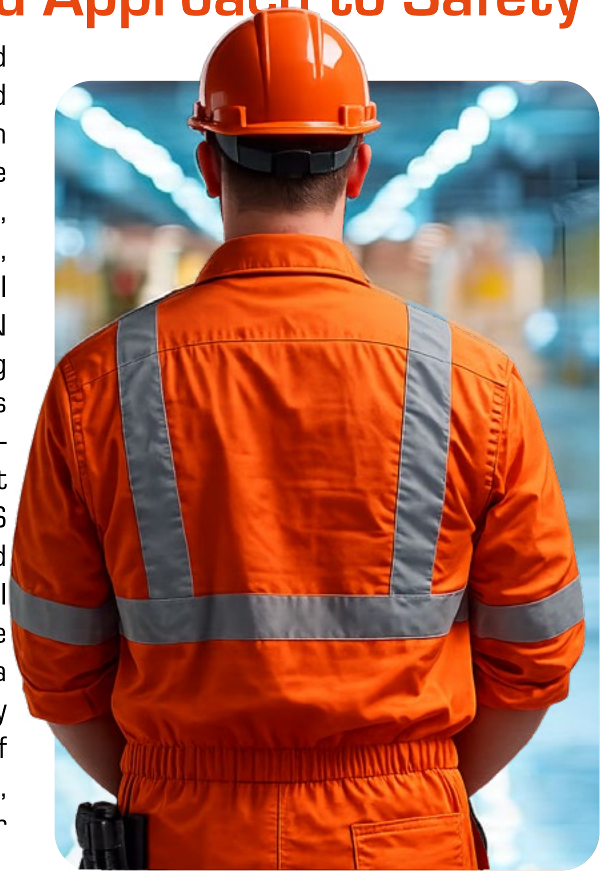
Hot Smoke Testing

One of the most advanced performance verification methods is the Hot Smoke Test. During this test, a controlled heat source—typically an alcohol-based burner or similar equipment—is used to simulate fire conditions and generate visible smoke. The test demonstrates not only smoke movement but also the behavior of hot smoke under realistic conditions, including its buoyancy and interaction with the jet fan system. Standards such as BS 7346-7 recommend this type of testing to validate overall system performance.

Periodic Maintenance and Long-Term Reliability

Smoke exhaust systems are largely dormant systems, spending most of their service life in standby mode. However, they must operate flawlessly when required. For this reason, regular self-testing routines and scheduled maintenance programs are essential. Routine activities such as bearing lubrication, damper seal inspections, and sensor calibration help ensure reliable performance throughout the system's operational life cycle.

Conclusion: An Integrated Approach to Safety



Smoke and heat exhaust systems for underground car parks are the result of a highly specialized engineering discipline. A successful system is not defined solely by powerful fans; it is the combination of accurate design and CFD analysis, high-performance smoke control dampers (Hound), intelligent automation systems (Hawk), and full compliance with international standards such as EN and NFPA. The evolving risk profile of modern parking facilities—driven by factors such as electric vehicles and increasingly constrained architectural spaces—has made jet fan technology an essential component of contemporary smoke control strategies. NOVVES solutions, including Dragonfly, Marlin, Hound, and Hawk, are designed to enhance both daily operational comfort and life safety performance during fire emergencies. Ultimately, smoke control serves as a building's safety assurance system. When properly designed, tested, and maintained, it becomes one of the most effective mechanisms for protecting lives, preserving property, and minimizing losses under the most challenging fire scenarios.

Why NOVVES?

- Certified products fully compliant with EN 12101-3 and NFPA 92 standards.
- System designs validated through CFD (Computational Fluid Dynamics) analysis.
- High-temperature-rated equipment certified to F300 (300°C / 2 hours) and F400 (400°C / 2 hours) classifications.
- Energy-efficient systems with CO/NOx sensor-based control.
- Class C airtightness-rated smoke dampers compliant with EN 1751.
- Solutions optimized for electric vehicle (EV) fire scenarios.
- Comprehensive commissioning, testing, and maintenance services.

Fire risks in enclosed and underground car parks continue to increase due to the growing use of plastics, synthetic materials, and lithium-ion batteries in modern vehicles. NOVVES jet fan–based smoke control systems provide an advanced engineering solution designed to address these evolving challenges while ensuring life safety, regulatory compliance, and operational reliability.

NOVVES

UNDERGROUND CAR PARKS

DRAGONFLY
Smoke and Heat Exhaust Fans

The Dragonfly series consists of F300 and F400 certified smoke and heat exhaust fans designed for operation under extreme fire conditions. Engineered to withstand temperatures of up to 300°C and 400°C for 2 hours, Dragonfly provides reliable performance for both emergency smoke extraction and daily ventilation applications.


Dragonfly T


Dragonfly JAU


Dragonfly R


Dragonfly JAR


Dragonfly JR


Dragonfly C

Dragonfly TJF
Tunnel Jet Fans

HOUND
Dampers

The Hound Series Dampers are designed to provide maximum control at critical points within ventilation and fire safety systems. Offering a range of solutions including shaft smoke dampers, ventilation louvers, and pressure relief dampers, the series stands out with its high airtightness, automation compatibility, and durability. By ensuring safe air management during fire emergencies, directing airflow in outdoor environments, and maintaining pressure balance within the system, Hound enhances both safety and operational efficiency in modern buildings.


Hound MSD


Hound MSFD-R


Hound MSFD-B

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1.2

ROAD AND RAIL TUNNEL SMOKE AND HEAT EXHAUST SYSTEMS

Evolution of Safety Requirements and Engineering Fundamentals

Tunnels and metro systems are indispensable components of modern transportation infrastructure. By their nature, they are enclosed underground environments with limited evacuation options. Heat, smoke, and toxic gases that would rapidly disperse in open air can become trapped within the tunnel geometry, turning the structure into a potentially life-threatening environment. In this context, ventilation systems are far more than climate-control equipment—they are among the most critical electromechanical systems within the infrastructure.

These fluid dynamics and thermodynamics-based systems are designed to:

During Normal Operation:
- Maintain air quality by removing emissions and heat loads.

During Emergency Conditions:
- Control smoke and create an aerodynamic shield that provides an **ASET (Available Safe Egress Time)** window for safe evacuation.

Strategic Objective:
- Prevent smoke back-layering through the critical velocity principle and provide a safe intervention corridor for emergency response teams.

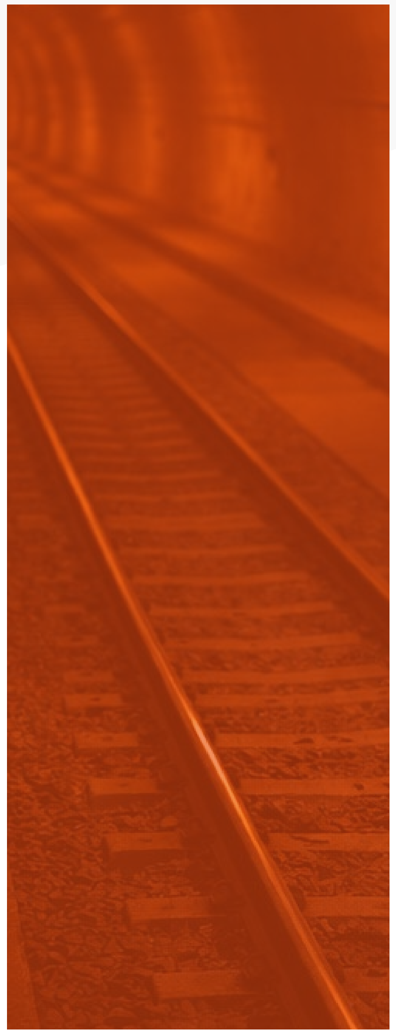
Tunnel Aerodynamics and Thermodynamic Forces

Tunnel ventilation systems are subject to continuously changing and non-linear dynamic forces that differ significantly from those encountered in buildings.

- **Piston Effect:** Generated by high-speed trains or heavy vehicles pushing air through the tunnel like a piston. Resulting pressure fluctuations of up to ±10 kPa can place significant mechanical stress on fans and dampers within seconds.
- **Stack Effect:** Caused by elevation differences between portals and temperature gradients within the tunnel. Particularly in inclined tunnels, the natural tendency of heated air to rise can create substantial resistance against the mechanical ventilation system.
- **Portal Wind Pressures:** Result from geographical wind conditions at tunnel entrances. When calculating fan thrust requirements, the most severe adverse wind scenario must be considered as part of the design process.

Key Engineering Design Principles

- **Critical Velocity Calculations:** The minimum airflow velocity required to prevent smoke from moving against the intended ventilation direction must be accurately determined.
- **Dynamic Control:** Automation systems should use real-time sensor data to continuously compensate for piston and stack effects, ensuring stable ventilation performance.
- **Zoning and Smoke Management:** During a fire event, smoke should be confined to a designated zone and directed away from evacuation routes and emergency access corridors.



Daily Operation, Emissions and Air Quality Management

The primary objective in road tunnels is the dilution and removal of CO, NO_2, and diesel particulate matter (DPM). While these pollutants are directly hazardous to human health, particulate matter can also reduce visibility and compromise operational safety.

- **Intelligent Control:** Optical and electrochemical sensors continuously monitor air quality and transmit real-time data to the SCADA system.
- **Energy Efficiency:** Variable Speed Drives (VSDs) dynamically adjust fan speeds according to pollution levels, reducing energy consumption and extending equipment service life.
- **Critical Scenario:** While the piston effect generated by moving traffic can assist ventilation under free-flow conditions, stop-and-go traffic places the full ventilation demand on the mechanical system.

Thermal Load and Comfort in Rail Systems

In metro and rail tunnels, the primary challenge is not emissions but the accumulation of heat. Braking friction, traction motors, and passenger-generated heat can significantly increase tunnel temperatures. To manage these thermal loads, localized ventilation strategies are commonly employed:

- **UPE (Under Platform Exhaust):** Extracts heat generated by braking systems and train undercarriage equipment directly at the source.
- **OTE (Over Track Exhaust):** Removes hot air discharged from train-mounted HVAC systems and provides smoke extraction capability during fire emergencies.

"These systems are designed to capture and extract heat at its source, preventing it from reaching passenger areas.

Fire Dynamics and Thermodynamic Behavior

Tunnel fires are significantly more aggressive than fires in open environments. As smoke reaches the tunnel ceiling, it spreads longitudinally, while radiant heat reflected from tunnel surfaces accelerates the risk of flashover.

- **Aerodynamic Capacity:** The primary design reference is the Heat Release Rate (HRR), which defines the size and intensity of the fire scenario.

Heat Release Rates (HRR) and Vehicle Categories

System capacity is determined based on the expected vehicle types using the following **MW (megawatt)** design fire values.

Vehicle Type	Heat Release Rate (HRR)	Fire Characteristics
Passenger Cars	5 MW	Rapid fire growth, low thermal load
Bus / Light Truck	15 - 20 MW	Moderate thermal load
Heavy Goods Vehicle (HGV)	30 - 50 MW	Very high heat release and dense smoke
Tanker / Hazardous Materials Vehicle	100+ MW	Extreme thermodynamic risk
Metro Train (Rail System)	10 - 30 MW	Rapid smoke spread in enclo-sed environments, prolonged thermal load

These values reflect the design criteria and fire scenarios accepted by international authorities and standards, including EN, NFPA, and PIARC.

Critical Success Factors

- **Material Resistance:** Exhaust ducts and fans must be certified for high-temperature operation, such as F250, F300, or F400 classifications (1 h / 2 h).
- **ASET Strategy:** The Available Safe Egress Time (ASET) must always exceed the time required for smoke conditions to become untenable.
- **Automation Integration:** Upon fire detection, the system must automatically switch to Emergency Mode within seconds and direct smoke toward the designated extraction route.

Fire Dynamics and Thermal Buoyancy

The most critical hazard in tunnel fires is the movement of smoke along the ceiling against the direction of fresh airflow. This phenomenon can:

- Fill evacuation routes with toxic gases such as CO and hydrogen cyanide (HCN).
- Prevent firefighting teams from reaching the seat of the fire.
- Create a strong thermal buoyancy force, which must be overcome by mechanical ventilation.

Critical Velocity Theory

At the core of tunnel ventilation engineering lies the challenge of overcoming the resistance created by hot smoke against the incoming fresh airflow. In a longitudinal ventilation system, mechanical fans generate a unidirectional airflow that pushes smoke toward the downstream portal.

However, due to its significantly lower density, hot smoke develops substantial thermal buoyancy and rises rapidly toward the tunnel ceiling. If the momentum generated by the jet fan system is insufficient to overcome this buoyancy force, hot smoke and toxic gases accumulated beneath the ceiling begin to travel upstream, moving against the ventilation direction toward the airflow source and evacuation routes. This hazardous reverse smoke movement is known as back-layering.

- **Calculation Method:** Critical velocity is calculated using models based on the Froude number, taking into account fire size (Q), tunnel height (H), cross-sectional area (A), and smoke temperature.
- **Gradient Effect:** In downhill tunnels, the natural upward movement of smoke increases ventilation demand, requiring greater airflow momentum and fan thrust.
- **Blockage Ratio:** In metro tunnel fire scenarios, a train may occupy 30–50% of the tunnel cross-sectional area. Therefore, calculations should be based on the net free area, excluding the space occupied by the train.

System Comparison Table

Ventilation System Type	Design & Operating Principle	Primary Applications & Advantages	Critical Disadvantages & Risks
Longitudinal Ventilation	Jet fans drive air in a single direction and exhaust smoke through tunnel portals.	Low construction cost. (Suitable for 3–5 km one-way tunnels)	Smoke spreads throughout the tunnel; hazardous in bidirectional traffic conditions.
Transverse Ventilation	Localized supply and exhaust are provided through parallel ducts and dampers.	Confines smoke at its source without allowing it to spread. (Common in metro systems and tunnels longer than 5 km)	Very high investment cost and requires a large tunnel profile.
Semi-Transverse Ventilation	Fresh air is supplied through ducts, while smoke is extracted through portals.	Optimizes air distribution.	Aerodynamic balancing is complex.

Advanced Engineering Solutions

Tunnel ventilation strategies and fan layout topologies vary depending on tunnel length, traffic direction (uni- or bi-directional), clearance requirements, and the design fire load. Selecting an inappropriate ventilation topology can directly increase life-safety risks during emergency situations. Saccardo nozzles and intermediate ventilation shafts are hybrid solutions used when tunnel length exceeds the effective range of conventional longitudinal ventilation. By introducing high-velocity air jets from outside the tunnel, these systems enhance aerodynamic performance and support smoke control. Intermediate shafts help maintain tunnel air velocities below 11.0 m/s (2200 fpm) by exhausting contaminated air and supplying fresh air where required.

- **Saccardo Nozzles:** Used in long tunnels to inject high-velocity air and enhance longitudinal airflow performance.
- **Intermediate Ventilation Shafts:** Remove contaminated air and provide fresh air supply, helping maintain airflow velocities below the comfort limit of 11.0 m/s.
- **Safety Standards:** All systems should comply with EN 12101 and NFPA 502 requirements and be capable of switching to full-capacity Emergency Mode within seconds during fire events.

UPE and OTE Systems in Emergency Response for Metro Stations

Ventilation systems in rail transit networks require dynamic response based on the train's location. In the event of a fire on a station platform, OTE (Over Track Exhaust) systems switch to emergency mode to prevent smoke from descending into passenger areas during evacuation. In this mode, smoke is extracted rapidly through high-level exhaust ducts at the station ceiling.

For tunnel fires occurring between stations, TVF (Tunnel Ventilation Fans) operate in a bidirectional Push-Pull configuration. One station supplies fresh air into the tunnel (Push), while the opposite station extracts smoke (Pull). This arrangement establishes the required critical velocity at track level, prevents back-layering, and creates a safe evacuation corridor.

Validation of Engineering Design: SES and CFD Simulation Techniques

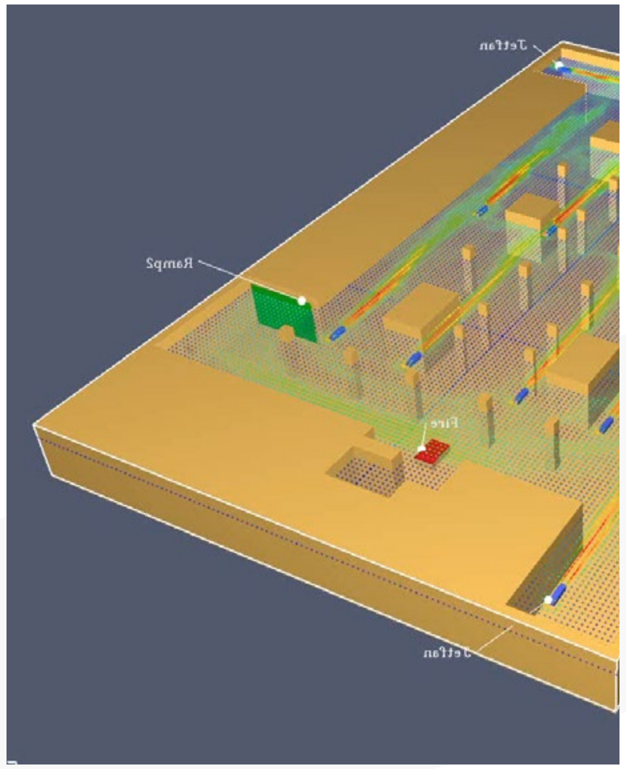
In underground infrastructure projects, assumptions are not sufficient; system performance must be verified through advanced simulation tools before construction begins. This validation process is primarily based on two complementary engineering disciplines.

Engineering and Design Verification

SES (Subway Environment Simulation) Analysis (1D Modeling)

SES (Subway Environment Simulation) software evaluates the entire underground network—including tunnels, shafts, stations, and ventilation systems—as an integrated thermodynamic model. It analyzes train-induced piston effects, large-scale pressure balances, and long-term heat accumulation throughout the network. Fan operating modes, airflow interactions, and potential short-circuit airflow paths are assessed during this stage. SES simulations are used to develop and validate emergency ventilation scenarios before detailed design implementation.

CFD (Computational Fluid Dynamics) Analysis (3D Modeling)



Using the macro-scale data generated by SES, the fire scenario is analyzed in three dimensions with CFD-based tools such as FDS at the specific ignition location. The CFD model divides the tunnel geometry into millions of computational cells and solves the Navier–Stokes equations for airflow, heat transfer, and smoke movement within each cell. Through these analyses, engineers can visualize temperature contours, airflow vectors, and toxic gas concentrations over time. The most critical evaluation criteria are measured at occupant level (1.8–2.0 m above floor level).

At this stage, the analysis verifies:

- Whether visibility remains above **10 meters,**
- Whether temperatures remain below the **60°C** tenability limit,
- Whether the Required Safe Egress Time **(RSET)** is shorter than the Available Safe Egress Time **(ASET)**.

International Safety Standards, Regulations and Egress Dynamics

All underground transportation infrastructure is designed in accordance with internationally recognized life-safety standards and regulations. The most widely adopted requirements are published by **NFPA (National Fire Protection Association)**. These standards define strict criteria covering fan temperature ratings, damper response times, tunnel lighting levels, smoke control performance, and the aerodynamic conditions of evacuation routes. A tunnel ventilation design is considered valid only when engineering calculations, CFD simulation results, and regulatory requirements are fully aligned with these international standards.

NFPA (130, 502) and European Standards (EN 45545, EU 2004/54/EC) NFPA 130 (Standard for Fixed Guideway Transit and Passenger Rail Systems)

NFPA 130 is the primary reference standard for rail transit system design in North America. It requires mechanical ventilation systems in enclosed tunnels exceeding 300 meters (1,000 ft) in length. To ensure safe evacuation, the maximum distance between stations is limited to 762 meters (2,500 ft), while the maximum spacing between cross passages is 244 meters (800 ft). The standard also defines minimum clear widths for emergency egress routes, including 610 mm (24 in.) for tunnel walkways, 813 mm (32 in.) for cross-passage doors, and 1,120 mm (44 in.) for station exit corridors. These requirements are based on passenger movement rates and evacuation time (RSET) calculations, including accessibility considerations.

NFPA 502 (Road Tunnels, Bridges, and Other Limited Access Highways)

For road tunnels, European regulations such as Directive 2004/54/EC and PIARC guidelines are applied alongside American standards. In particular, tunnels exceeding 1,000 meters generally require mechanical smoke extraction systems. Critical velocity criteria and back-layering control methodologies are further defined using empirical data obtained from full-scale fire testing.

Performance Testing and Equipment Certification

Regardless of how advanced a design may be, engineering assumptions must be validated under real fire and operational conditions through independent accredited testing laboratories (such as ISO/IEC 17025 accredited facilities). For underground transportation applications, certification is typically based on two primary testing categories.

Aerodynamic Performance and Acoustic Testing (AMCA and ISO Integration)

In North America, fan aerodynamic performance is measured according to AMCA 210, while its international equivalent is ISO 5801. Tunnel jet fan thrust performance is tested in accordance with AMCA 250 and the equivalent ISO 13350 standard. Noise and sound power measurements are certified under AMCA 300/320 and their international equivalents, including ISO 13347 (and ISO 5136 for in-duct measurements). These tests quantify and certify the fan's aerodynamic thrust force (measured in Newtons) under controlled laboratory conditions.

High Temperature and Fire Resistance (EN 12101-3)

Compliance with EN 12101-3 is essential for smoke exhaust systems operating during fire conditions. Fans certified to F300/F400 standards are tested to operate continuously for 2 hours at 300°C or 400°C. Upon successful testing, products are awarded the CE mark. In addition to certification, large-scale projects often require Factory Acceptance Tests (FAT) to verify performance, noise, and vibration levels before equipment is delivered to site.

Integration of Technological Components and the NOVVES System Ecosystem

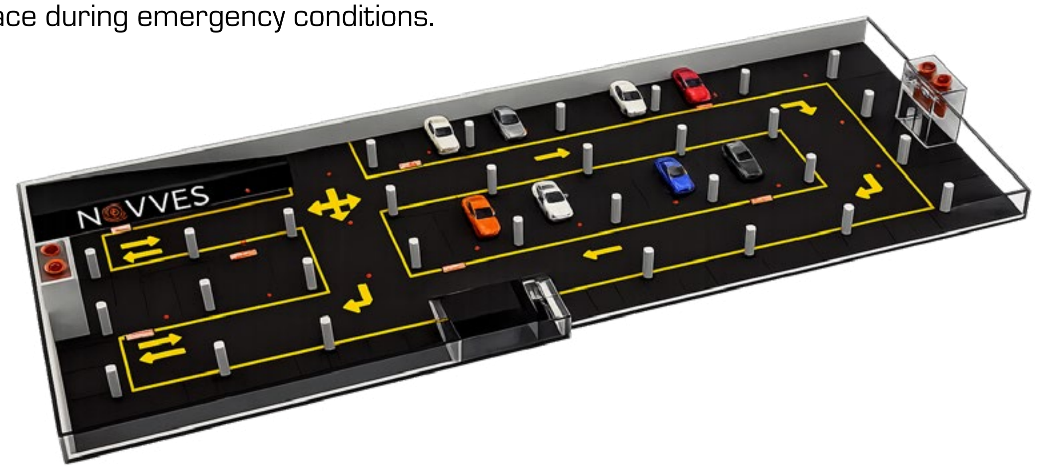
Underground infrastructure equipment differs significantly from conventional building products, as it must operate under extreme pressure, corrosive environments, and temperatures exceeding 300°C–400°C. Developed with a zero-failure philosophy, the NOVVES ecosystem (Dragonfly, Hound, Marlin, and Hawk) delivers a fully integrated solution architecture designed to perform reliably under the most demanding conditions.

Dragonfly J Series (Tunnel Jet Fans)

The Dragonfly J Series is engineered to generate high thrust in restricted tunnel clearances. Its key engineering advantage is a 100% reversible airflow capability, allowing airflow direction to be changed within seconds according to the fire scenario. The rotor geometry produces symmetrical thrust in both directions, minimizing momentum losses. F300/F400-certified casings and Class H insulated motors continue supporting smoke control operations even under extreme fire conditions.

Dragonfly TAA Series (Large Axial Main Shaft Fans)

The Dragonfly TAA Series consists of high-capacity axial fans designed for operation in OTE/UPE systems and major ventilation shafts within metro stations and underground transportation facilities. Equipped with impellers up to 3 meters in diameter, these units are capable of overcoming significant static pressure losses and extracting large volumes of smoke and hazardous gases to the surface during emergency conditions.

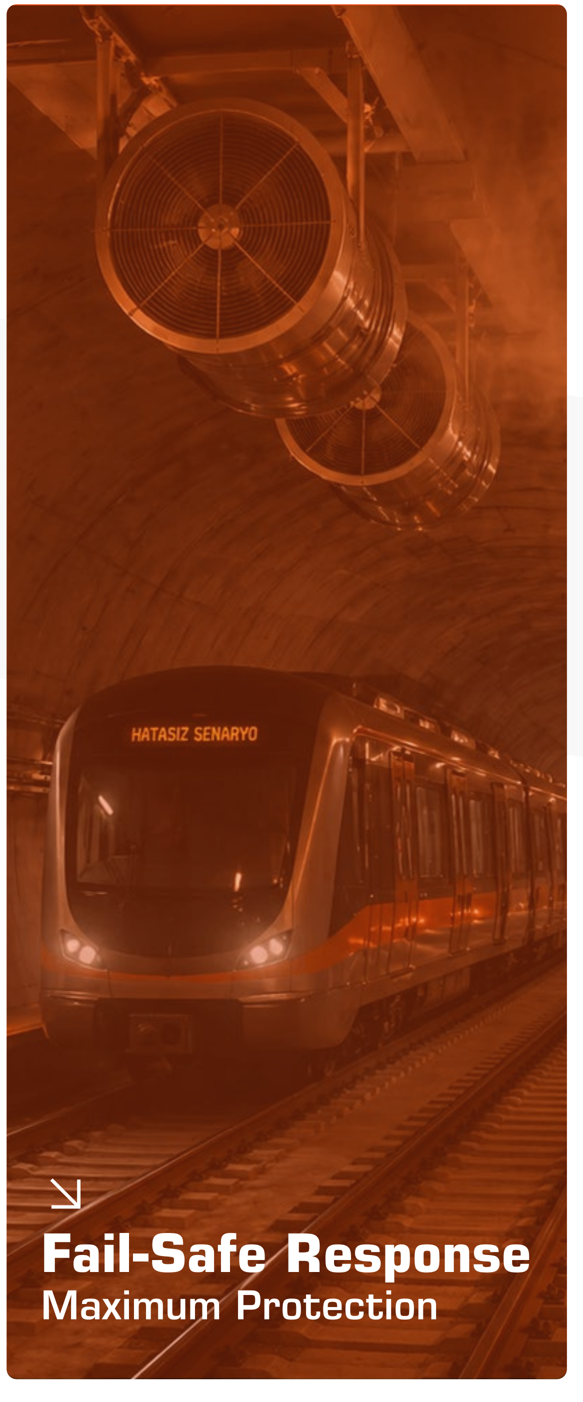


Hound Series: High-Pressure Smoke Control

The Hound Series heavy-duty smoke dampers are engineered to withstand the high-pressure fluctuations generated by moving trains while operating reliably under 300°C/400°C fire conditions. Designed with precision engineering, they resist deformation under extreme heat and, together with certified high-leakage-class fire-resistant seals, create a robust barrier that prevents smoke migration into clean shafts and protected areas.

Marlin Series: Pressurization and Safe Escape Protection

During a fire event, keeping escape routes smoke-free is just as critical as extracting smoke. The Marlin Series pressurization fans create positive pressure in stairwells, shelters, and protected escape zones, forming an invisible air barrier that prevents smoke infiltration and ensures safe evacuation.



Hawk Series: Central Intelligence, SCADA and Autonomous Response

The true capability of a smoke control system emerges when mechanical equipment is integrated with a centralized control platform such as Hawk SCADA and PLC Automation. Data collected from temperature, gas, and airflow sensors distributed throughout the tunnel is processed by redundant controllers designed to meet high safety integrity requirements, including SIL 2/3. The system identifies the exact fire location (zone) and automatically selects the most appropriate emergency scenario from a library of pre-engineered response strategies. Within seconds, Hawk determines which fans should reverse direction, which dampers should open or close, and how airflow should be managed. Variable Speed Drives (VSDs) ensure smooth and reliable motor control while precisely regulating ventilation performance according to the scale and development of the fire event.

In Brief

Smoke and heat exhaust systems in road tunnels and underground metro networks are not optional mechanical systems—they are fundamental elements of the infrastructure itself. In environments where thousands of people travel every day and critical transportation links must remain operational, ventilation becomes a life-safety function rather than a comfort feature.

Under normal operating conditions, these systems manage vehicle emissions and maintain air quality. During a fire, however, the challenge becomes far greater. Temperatures can rise above 1,000°C within minutes, and failure to achieve the required critical velocity may result in back-layering, allowing smoke and toxic gases to spread into evacuation routes.

Successfully controlling these conditions requires a combination of international standards such as NFPA 130 and NFPA 502, advanced SES network modeling, and detailed CFD simulations. Yet engineering calculations alone are not enough. Reliable performance depends on certified equipment tested to standards such as EN 12101-3 (F300/F400) and validated through rigorous aerodynamic and fire-performance testing.

With Dragonfly tunnel fans, Hound smoke dampers, Marlin pressurization units, and Hawk automation systems, NOVVES delivers a fully integrated smoke control ecosystem. Together, these technologies create an intelligent, tested, and highly reliable safety solution for the world's most demanding underground transportation projects.

> **In underground engineering, technology is more than infrastructure, it is the silent guardian that transforms science into life safety when every second matters.**

DRAGONFLY
Smoke and Heat Exhaust Fans

The Dragonfly series consists of F300 and F400 certified smoke and heat exhaust fans designed for operation under extreme fire conditions. Engineered to withstand temperatures of up to 300°C and 400°C for 2 hours, Dragonfly provides reliable performance for both emergency smoke extraction and daily ventilation applications.


Dragonfly T


Dragonfly JAU


Dragonfly R


Dragonfly JAR

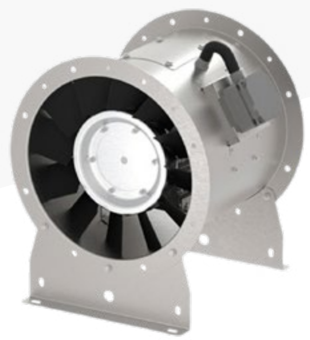

Dragonfly JR


Dragonfly C

Dragonfly TJF
Tunnel Jet Fans

MARLIN
Axial Fans

The Marlin Series tubular axial fans are designed for a wide range of ventilation applications, including fresh air supply, exhaust ventilation, and duct-mounted operation. In fire scenarios, they can also function as pressurization fans, preventing smoke from entering escape routes such as stairwells and elevator shafts. Thanks to their compact design, Marlin fans can be installed directly within duct systems, shafts, or rooftop applications. Combining high performance with low noise levels, they provide efficient and comfortable ventilation in a wide variety of environments.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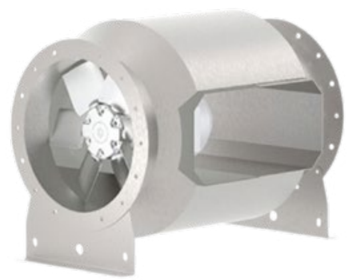
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SHOPPING MALLS

Shopping malls are large-scale, high-ceiling structures with significant occupant density and complex multi-level layouts. In such environments, natural air movement is insufficient. During a fire event, smoke can rapidly accumulate at upper levels, reducing visibility, triggering panic, and obstructing safe evacuation. Therefore, ventilation systems in shopping malls are designed not only for occupant comfort, but also to control smoke and support life safety during emergencies.

Fire Safety and Smoke Management Fundamentals in Complex Commercial Buildings

Modern shopping malls feature vast atriums, interconnected gallery spaces, extensive parking areas, and the capacity to accommodate thousands of occupants simultaneously. While these architectural characteristics enhance functionality, they also introduce significant fire safety challenges. Engineering studies and fire statistics consistently show that the primary cause of fire-related fatalities is not direct flame exposure, but the inhalation of smoke containing carbon monoxide (CO) and other toxic gases, as well as the severe loss of visibility caused by smoke accumulation. For this reason, smoke and heat exhaust systems installed in shopping malls are far more than conventional ventilation solutions. They represent one of the most critical life-safety infrastructures within the building, specifically engineered to control smoke movement, maintain tenable conditions, and facilitate the safe evacuation of occupants during fire emergencies.

Atrium Aerodynamics and the “Plugholing” Phenomenon

In large-volume spaces such as shopping mall atriums, smoke extraction relies on the principle that smoke generated during a fire rises rapidly due to thermal buoyancy and forms a smoke layer beneath the ceiling. The primary engineering objective is to maintain this smoke layer above occupant level. One of the most critical risks in this process is plugholing, which occurs when exhaust fans penetrate the smoke layer and begin extracting clean air from below instead of smoke. When this happens, fan capacity is wasted on clean air, while smoke remains trapped and may descend into occupied areas. To prevent this phenomenon, systems are optimized through CFD analysis using multiple low-flow extraction points (multi-point extraction) rather than a single high-capacity exhaust location.

Smoke Zoning and Fire / Smoke Curtains

- To prevent uncontrolled smoke spread in large commercial buildings, the structure is divided into dedicated smoke zones and reservoirs using smoke curtains. During a fire, these curtains descend from the ceiling and act as barriers, confining smoke within a designated area and preventing horizontal smoke migration.
- By concentrating smoke near its source, extraction fans or smoke vents located above can remove it far more efficiently. This controlled smoke management strategy extends available evacuation time and helps firefighters localize the incident area. Without such compartmentation, attempting to extract smoke from an unrestricted smoke mass across an entire shopping mall becomes technically impractical.

Ventilation and Jet Fan Technology in Underground Car Parks

Large underground parking facilities, an integral part of modern shopping mall infrastructure, are increasingly ventilated using Jet Fan technology, adapted from tunnel ventilation engineering, instead of traditional ducted systems. The dead zones, ceiling height losses, and high installation costs associated with ducted ventilation have been overcome through the momentum-based operating principle of jet fan systems. Jet fan systems serve two distinct operational functions:

- **Daily Operation (CO Extraction):** Toxic carbon monoxide (CO) and other vehicle exhaust gases are diluted and removed based on data transmitted from CO detectors to the SCADA control system. The automation system operates only the fans in areas where CO concentrations increase, providing significant energy savings.
- **Fire Emergency (Smoke Control):** When a fire is detected, jet fans switch to emergency mode and rapidly drive smoke toward the main exhaust shafts.
- Unlike conventional ventilation systems, jet fan performance is evaluated not by airflow rate alone, but by thrust force (Newton), as in tunnel ventilation applications.

Here, thrust force (Newton) is determined by air density (kg/m^3), discharge velocity (m/s), and fan outlet area (m^2). The high-velocity air jet generated by a jet fan entrains the surrounding stationary air (induction effect), mobilizing an air mass approximately 8 to 16 times greater than the fan's own airflow.

Validation of Engineering Design: CFD Analysis (AMCA and ISO Integration)

In complex environments such as shopping malls, where retail areas, atriums, and parking structures are interconnected, fire dynamics cannot be accurately predicted using simple calculations. Design validation must be performed in advance through Computational Fluid Dynamics (CFD) simulation software such as FDS or ANSYS Fluent. CFD analysis divides the shopping mall geometry into millions of three-dimensional computational cells (mesh elements). A fire with a defined Heat Release Rate (HRR) is then simulated, allowing engineers to evaluate smoke propagation, airflow patterns, plugholing risks, and thermal stratification in real time. Most importantly, the analysis verifies whether visibility at occupant level remains above 10 meters and whether environmental temperatures stay within accepted tenability limits during evacuation. Based on these results, fan capacities can be increased and equipment layouts optimized to ensure compliance with life-safety objectives.

International Safety Standards and Regulations

Smoke control systems for shopping malls and parking structures must comply with globally recognized life-safety standards. Key regulations include:

- **NFPA 92 (Standard for Smoke Control Systems):** The primary design standard for smoke management in shopping malls and atriums in North America. In Europe, smoke barriers and zoning are governed by EN 12101-1, while mechanical smoke exhaust calculations and plugholing analyses are performed in accordance with EN 12101-5. Where Natural Smoke and Heat Exhaust Ventilators (NSHEVs) are used, the products must comply with EN 12101-2
- **BS 7346-7:** A British standard requiring a minimum of 10 air changes per hour in large enclosed car parks (typically exceeding 2,000 m²) during fire conditions and recommending separate smoke extraction routes for each fire zone.
- **EN 12101-3:** The European performance standard certifying that jet fans and atrium exhaust fans can continue operating under elevated temperatures during a fire event (e.g., F300 – 300°C for 2 hours; F400 – 400°C for 2 hours).

Technological Components and Integration of the NOVVES System Ecosystem

Shopping mall smoke control systems consist of high-temperature-resistant motors, fast-acting smoke-tight dampers, and intelligent automation panels that manage the entire network in real time. NOVVES provides a fully certified product ecosystem engineered for the demanding conditions of shopping malls, from enclosed parking garages to large atriums.

- **Dragonfly Series (Smoke & Heat Exhaust and Jet Fans):** EN 12101-3 F300/F400 certified jet fans that dilute carbon monoxide in daily operation and create thrust-driven airflow barriers during fire events. The series also includes large axial exhaust fans used in atrium smoke extraction shafts.
- **HOUND Series (Heavy-Duty Dampers):** Automatic dampers designed to maintain pressure balance and prevent smoke migration between zones. They feature high-leakage-class performance, aerodynamic design, and reliable operation under emergency conditions.
- **MARLIN and TURTLE Series:** High-efficiency pressurization and axial fans used to protect stairwells, elevator shafts, and escape corridors from smoke infiltration while also supporting general ventilation requirements.

Smoke management in shopping malls and parking facilities is an engineering-driven process governed by NFPA 92 and supported by intelligent jet fan technology. System performance depends on CFD-based validation, high-temperature-resistant equipment, and seamless automation integration, ensuring that smoke is effectively controlled and removed while maintaining occupant safety.

SHOPPING MALLS

DRAGONFLY
Smoke and Heat Exhaust Fans

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Dragonfly T

Dragonfly JAU

Dragonfly R

Dragonfly JAR

Dragonfly JR

Dragonfly C

Dragonfly TJF
Tunnel Jet Fans

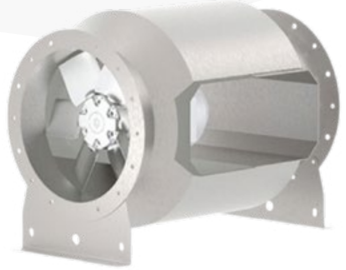
HOUND
Dampers

The Hound Series Dampers are designed to provide maximum control at critical points within ventilation and fire safety systems. Offering a range of solutions including shaft smoke dampers, ventilation louvers, and pressure relief dampers, the series stands out with its high airtightness, automation compatibility, and durability. By ensuring safe air management during fire emergencies, directing airflow in outdoor environments, and maintaining pressure balance within the system, Hound enhances both safety and operational efficiency in modern buildings.

Hound MSD

Hound MSFD-R

Hound MSFD-B



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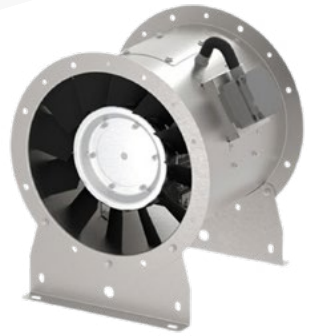

Marlin

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Axial Fans

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STADIUMS AND ASSEMBLY HALLS

Points to Consider

- Maintaining smoke under control at the upper level
- Protecting escape routes from smoke exposure
- Planning fan locations according to building volume and fire scenarios
- Designing airflow direction to support evacuation paths
- Dynamic control through intelligent automation systems
- Compliance with international fire and ventilation standards

In stadiums and assembly halls, ventilation is not designed merely for air circulation; it is engineered to protect human life.

Smoke Extraction During Emergencies

In large-volume venues such as stadiums and convention centers, the primary hazard during a fire is toxic smoke that rapidly reduces visibility. The main engineering objective is to keep smoke away from evacuation routes throughout the required evacuation period (RSET) while maintaining acceptable tenability conditions. Due to varying fire loads and high occupant densities, these facilities present some of the most challenging smoke control scenarios in fire safety engineering.

Daily Operation and Comfort Management

The CO_2 levels and thermal loads generated by fluctuating occupant densities are managed through Demand-Controlled Ventilation (DCV) strategies. Using intelligent sensors and autonomous controls, the system provides maximum fresh air during peak occupancy and reduces fan speeds during partial occupancy to achieve significant energy savings. Jet fans and airflow control devices help eliminate dead zones, optimize air distribution, and maintain thermal comfort throughout the venue.

Smoke Management During Fire Events

Systems designed in accordance with NFPA standards promote smoke stratification by allowing smoke to rise and form a stable layer beneath the roof. Automatic smoke vents and high-capacity exhaust fans prevent smoke descent, keeping evacuation routes clear and accessible. As a result, smoke is maintained at safe levels, supporting effective life safety protection and orderly evacuation.

Make-up Air Supply and Plugholing Risk

Adequate make-up air is essential for effective smoke extraction. Without it, negative pressure can develop, preventing evacuation doors from opening and trapping occupants. In addition, if fan speeds are not properly optimized, plugholing may occur, causing exhaust fans to draw clean air instead of smoke and significantly reducing system effectiveness.

Smoke Extraction and Pressurization of Escape Routes

According to NFPA 101, stadiums require dedicated systems to protect occupants from smoke infiltration during evacuation. Escape routes, VIP areas, and stair towers are maintained under positive pressure (typically +50 Pa) using pressurization fans. This creates an aerodynamic barrier that prevents smoke from entering protected spaces and ensures safe egress conditions.

CFD-Based Design Validation

In complex structures, smoke behavior is verified through 3D CFD simulations. These analyses digitally evaluate smoke movement and fan performance, demonstrating that visibility levels and temperatures along evacuation routes remain within safe limits under fire conditions.

National and International Safety Standards

Smoke control is critical for safe evacuation in large-volume buildings. The following standards serve as key design references:

- **BYKHY (Turkish Fire Regulation):** Mechanical smoke extraction and 60-minute fire-resistant cabling are mandatory for buildings over 51.50 m in height or 1,000+ m² in area.
- **NFPA 92 / NFPA 101:** Primary guidelines for smoke control system design and maintaining tenable conditions during evacuation.
- **EN 12101:** Defines the high-temperature performance requirements for smoke exhaust fans and vents, including F300/F400 classifications.

Technological Components and the NOVVES System Ecosystem

The demanding ventilation requirements of stadiums and large assembly venues cannot be met with conventional HVAC equipment. NOVVES provides a fully integrated ecosystem of certified products and automation solutions, engineered specifically for large-scale public facilities.

In Summary,

Mechanical smoke extraction systems are essential in stadiums, arenas, and other large assembly buildings. Designed in accordance with EN 12101, NFPA 92/101, and BYKHY, these systems are validated through CFD simulations to ensure safe evacuation conditions. NOVVES solutions, including the Marlin, Turtle, and Dragonfly series, maintain airflow balance and create a reliable safety barrier for high-occupancy environments.

STADIUMS AND ASSEMBLY HALLS

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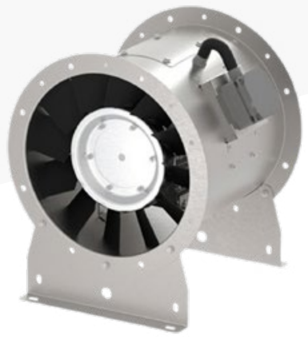

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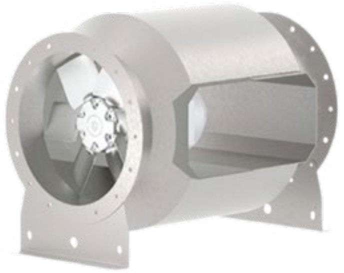

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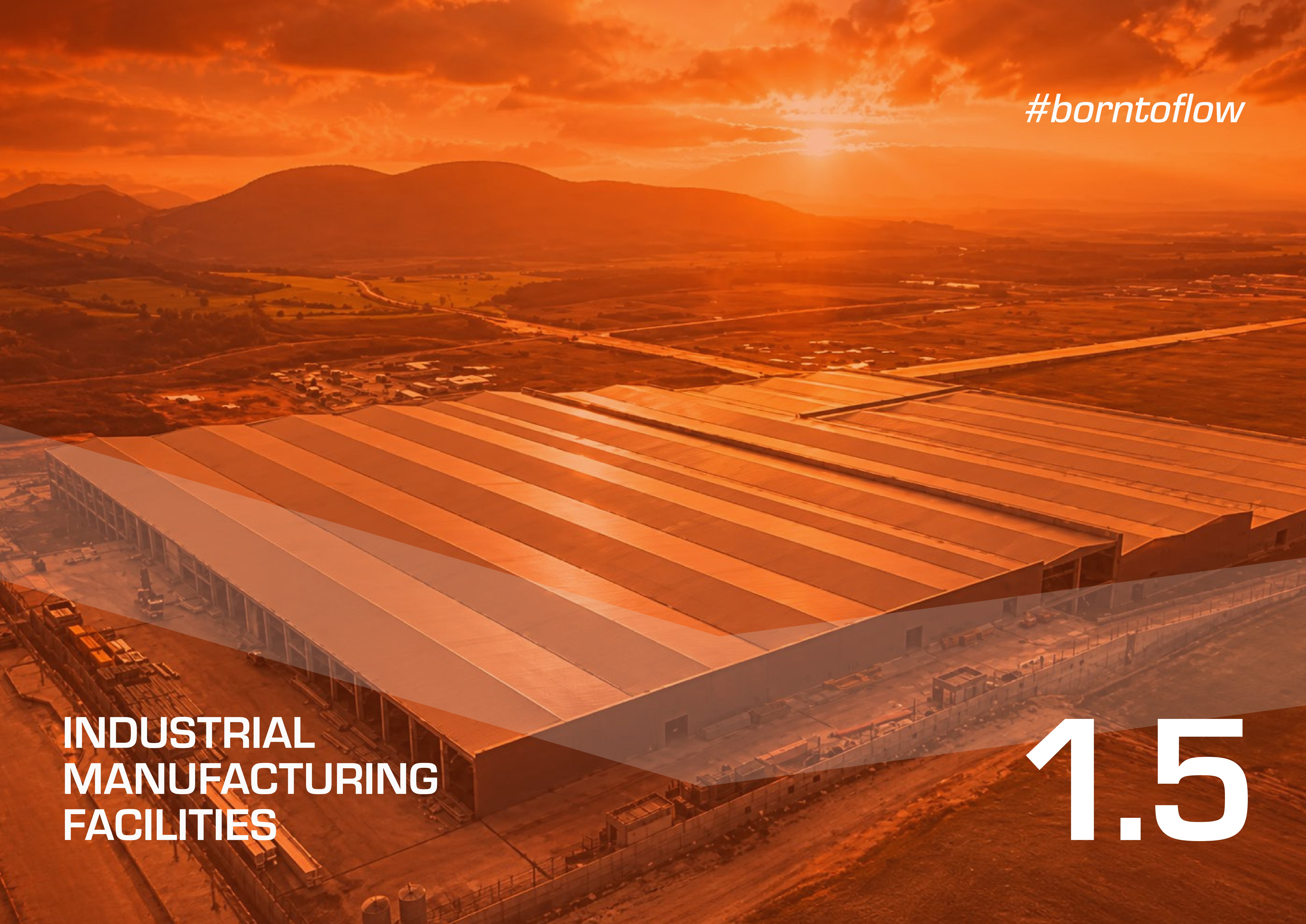

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INDUSTRIAL MANUFACTURING FACILITIES

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INDUSTRIAL MANUFACTURING FACILITIES

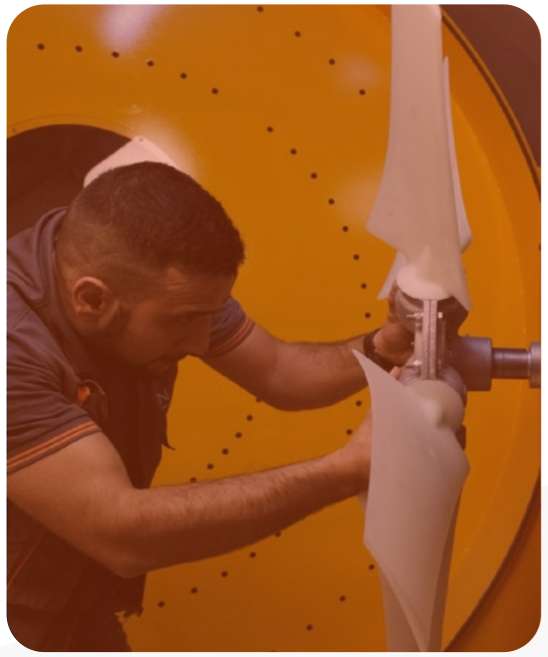
The Critical Role of Ventilation Engineering in Heavy Industry and Manufacturing Facilities

- Industrial manufacturing facilities—including automotive plants, metallurgical facilities, chemical and food processing plants, and foundries—are fundamentally different from commercial buildings designed primarily for occupant comfort. They are highly complex enclosed environments where high thermal loads, toxic emissions, and demanding industrial processes must be continuously managed. In these facilities, mechanical ventilation and smoke control systems perform vital functions such as extracting hazardous fumes and process gases, removing excess process heat, and protecting the facility's structural integrity during fire emergencies.
- Effective industrial ventilation engineering requires the management of smoke, heat, and process-generated contaminants. Depending on the nature of the production process, facilities may generate high thermal loads, toxic gases, solvent vapors, welding fumes, or combustible/explosive dust. These hazards must be assessed not only according to general ventilation principles but also in compliance with standards and regulations such as NFPA 91, NFPA 204, EN 12101, EN 16798-3, EN ISO 14123, EN 1093, EN ISO 21904, ATEX 2014/34/EU, and, where applicable, EN 14986. Accordingly, system design should incorporate principles of Local Exhaust Ventilation (LEV), general dilution ventilation, smoke and heat extraction, explosion protection, and control of process-related hazardous emissions.

“ In industrial manufacturing facilities, ventilation is designed not merely to cool the environment, but to ensure process safety and protect personnel from hazardous gases and airborne contaminants.

Process-Generated Hot Gas Extraction

Industrial manufacturing processes generate high-temperature gases, steam, oil mist, solvent fumes, and airborne contaminants. Welding lines, foundries, paint booths, heat-treatment lines, drying tunnels, and chemical processes can significantly increase ambient temperatures and directly impact workplace safety. Therefore, the controlled removal of hot and contaminated air is essential not only for employee health and safety but also for equipment reliability and service life.



Local Exhaust Ventilation (LEV), NFPA 91, and the European EN/ATEX Approach

In industrial facilities, contaminants such as welding fumes, solvent vapors, acid fumes, oil mist, dust, particles, and process emissions should be captured and removed directly at the source before they disperse throughout the facility. This principle is known as Local Exhaust Ventilation (LEV). LEV systems typically consist of hoods, extraction arms, process connection points, ductwork, filtration and separation units, air-cleaning equipment, heavy-duty industrial fans, and safe discharge stacks. The UK HSE recognizes LEV as the primary control method for capturing airborne contaminants before workers can inhale them.

In North America, one of the key references for the fire and explosion safety of such systems is NFPA 91 – Standard for Exhaust Systems for Air Conveying Vapors, Gases, Mists, and Particulate Solids. NFPA 91 establishes design and installation requirements aimed at reducing fire and explosion risks in exhaust systems handling vapors, gases, mists, smoke, and particulate matter.

Within the European framework, there is no direct equivalent to NFPA 91. Instead, process emission control is addressed through standards such as EN ISO 14123, which provides principles for reducing health risks from hazardous substances released by machinery; EN 1093, covering testing and measurement methods for airborne emissions; and EN ISO 21904, which specifically addresses welding fume extraction and control applications.

If the extracted smoke, vapor, or dust is combustible or explosive, the system must be evaluated not only as a ventilation system but also within the scope of ATEX safety requirements. In such cases, EN 1127-1 should be applied for explosive atmosphere risk assessment, while EN 14986 should be considered for the design and marking requirements of fans operating in explosive atmospheres. Selecting the correct duct air velocity is critical in these systems. Insufficient airflow may allow particles and contaminants to settle inside ductwork, leading to blockages, increased fire load, and reduced filter and fan performance. Therefore, LEV systems should be designed with delayed shutdown (purge) operation, differential pressure monitoring, minimum airflow control, and appropriate spark and static electricity protection measures for fire- and explosion-risk applications.

Smoke and Heat Management in Industrial Facilities (NFPA 204, EN 12101-2 / 3)

Smoke control is essential in manufacturing and logistics facilities with high fire loads. Systems designed in accordance with NFPA 204 and EN 12101 standards aim to maintain the smoke layer above occupant level. Make-up air supply replaces extracted smoke, preventing negative pressure and plugholing effects while ensuring safe evacuation conditions.

Corrosive, High-Temperature, and Abrasive Environment Dynamics

In highly corrosive environments, such as acidic process areas, durable plastic fans manufactured from polypropylene (PP) or polyethylene (PE) are commonly preferred. For high-temperature processes such as foundries and industrial furnaces, specialized bifurcated or belt-driven systems with thermal insulation and motors located outside the airflow are used, capable of continuous operation at temperatures of 200°C–300°C. These solutions provide reliable operational safety beyond standard fire-resistance requirements, ensuring continuous performance under demanding industrial conditions.

Technological Components and the Integration of the NOVVES System Ecosystem

Designed to meet the high-capacity and 24/7 operational demands of industrial facilities, the NOVVES ecosystem provides certified solutions tailored to each application. Nautilus centrifugal fans overcome the high resistance of filtration and duct systems in local exhaust applications, while the high-thrust Dragonfly series ensures effective smoke control during fire events. The wall-mounted Owl series supports general ventilation and smoke dilution, helping maintain the highest levels of facility safety.

Industrial manufacturing facilities are environments where mechanical ventilation evolves from a comfort system into a critical life-safety, smoke control, and process protection solution. From welding fumes to extreme process heat, each contaminant requires a dedicated ventilation strategy. While LEV systems designed in accordance with NFPA 91 capture hazardous emissions directly at the source, NFPA 204 principles help control smoke stratification in large industrial halls. NOVVES supports operational continuity and facility safety through its heavy-duty Nautilus centrifugal fans and F300/F400-certified Dragonfly product family, delivering reliable performance even under the most demanding smoke and heat scenarios.

Combining industrial power with advanced engineering, NOVVES delivers uncompromising safety even in the most demanding operating environments.

NAUTILUS
Industrial Fans

Nautilus Silence

High-performance industrial fans combine powerful airflow with energy efficiency through their backward-curved impeller design and direct-drive motor configuration. They are ideally suited for extracting air containing dust, chips, oil mist, and similar particulates, while also providing reliable fresh air supply in ventilation applications. Designed for quiet operation, these fans offer flexible airflow direction adjustment through their reversible housing configuration. Durable, efficient, and adaptable, they provide an ideal solution for demanding industrial environments.

Nautilus LFP

OWL
Wall-Mounted Fans

The Owl Series Wall-Mounted Fans offer easy direct-wall installation with both square and circular frame options. Designed for large spaces requiring high airflow rates—such as warehouses, hangars, paint shops, and manufacturing facilities—they provide efficient air extraction and ventilation performance. Engineered for long service life and minimal maintenance, the Owl Series delivers maximum airflow and efficiency through its compact design and optimized blade geometry while maintaining low noise levels. Combining durability, energy efficiency, and reliable performance, it is an ideal solution for professionals seeking effective ventilation in demanding industrial environments.

Owl RWA

Owl RER

Owl CER

HERON
Roof Fans

The Heron Series Roof Fans are designed for rooftop installation to improve indoor air quality and are primarily used as exhaust fans. They provide safe and efficient extraction of air to the outdoor environment. Upon request, the Heron AH Series can also be manufactured as a supply air unit, delivering fresh air into the building while maintaining the same reliable performance and durability.

Heron RH

Heron AH

Heron RHS

Heron RV

Heron RVS

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1.6

HOSPITALS AND HEALTHCARE FACILITIES

HOSPITALS AND HEALTHCARE FACILITIES

Fundamentals of Fire Safety in Healthcare Facilities

Hospitals represent some of the most demanding environments in terms of smoke management and life safety engineering. Safety in these facilities is ensured through complex electromechanical infrastructures designed in accordance with international standards such as NFPA, HTM, and EN 13501. The primary objective of mechanical smoke extraction systems is to prevent smoke migration into critical hospital areas while maintaining tenable conditions along evacuation routes. These vital systems not only provide valuable time for patient evacuation but also create safe access corridors for firefighting operations.

- Smoke and Toxic Gas Control
- Compliance with International Standards
- Fire-Resistant Material Selection
- Preservation of Tenability Conditions
- Firefighter Access and Intervention Routes

Defend-in-Place Strategy and the Total Concept

Hospitals are among the most critical buildings where conventional full-building evacuation is often impractical due to patients with limited mobility. Instead of evacuating the entire facility, the Defend-in-Place strategy focuses on relocating patients horizontally into protected smoke compartments and safe zones. Defined by NFPA 101, this approach relies on effective compartmentation, smoke barriers, and fully integrated fire protection systems, ensuring patient safety without requiring complete building evacuation.

Smoke Zoning, HVAC Automation, and Pressurization Systems

The success of the Defend-in-Place strategy in hospitals depends on the integrated operation of mechanical smoke control and HVAC automation systems. In the United States, these systems are designed in accordance with NFPA 92, while in Europe they follow the EN 12101 series and EN 54 fire detection and alarm standards. During a fire event, the system coordinates smoke dampers, smoke control fans, pressurization fans, and air handling units to limit smoke migration beyond the affected fire compartment. Escape routes, stairwells, corridors, and critical areas are protected against smoke infiltration by maintaining positive pressure through differential pressure systems defined under EN 12101-13. This ensures the protection of horizontal evacuation routes and safe refuge areas for patients.

For reliable operation, the system must be activated not only through manual controls but also through automatic fire detection systems compliant with the EN 54 series, including addressable detectors, sprinkler flow switches, pressure sensors, and damper position feedback devices. Smoke control fans should comply with EN 12101-3, smoke control ducts with EN 12101-7, smoke control dampers with EN 12101-8, and power supply systems with EN 12101-10.

Smoke Management in Hospital Atriums and Large-Volume Spaces

Smoke management in hospital atriums and large-volume spaces must be evaluated in accordance with NFPA 92 as well as the EN 12101 series in Europe. For atriums and complex buildings, CEN/TR 12101-5 provides design guidance for smoke and heat exhaust calculations, while EN 12101-3 covers mechanical smoke exhaust fans, EN 12101-7 smoke control ductwork, EN 12101-8 smoke control dampers, and EN 12101-10 power supply systems. These systems are designed to prevent the smoke layer from descending below safe occupant levels, ensuring tenable conditions for evacuation and firefighting operations. NOVVES provides solutions focused on high reliability, airtightness, and high-temperature resistance for these critical applications.

DRAGONFLY

Smoke and Heat Exhaust Fans

Dragonfly Series is a powerful smoke extraction solution specifically designed for high-temperature fire conditions and certified to F300/F400 fire resistance standards. Engineered to operate continuously for 2 hours at temperatures of 300°C and 400°C, these fans ensure reliable smoke and toxic gas extraction during fire emergencies. In addition to fire applications, Dragonfly fans are widely used in enclosed parking garages and similar environments for the extraction of contaminated air containing carbon monoxide (CO) and hydrocarbons, providing safe and efficient ventilation under demanding conditions.

Dragonfly T

Dragonfly JAU

Dragonfly R

HOUND

Dampers

Hound Series Dampers are designed to provide maximum control at critical points within ventilation and fire safety systems. Offering a range of solutions including shaft smoke dampers, weather louvers, and pressure relief dampers, the series stands out with its high airtightness, automation compatibility, and durability. By ensuring safe smoke extraction during fire events, directing airflow in outdoor applications, and maintaining pressure balance within ventilation systems, Hound dampers enhance both safety and operational efficiency in modern buildings.

Hound MSD

Hound MSFD-R

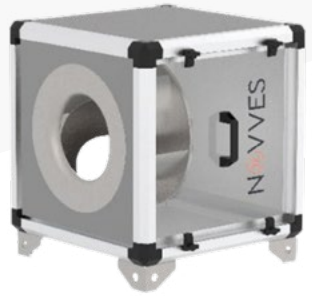
TURTLE

Cabinet Fans

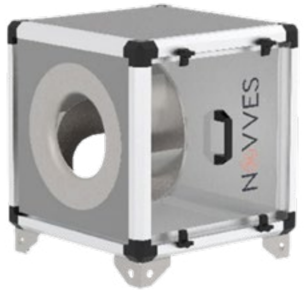
Turtle Series Cabinet Fans are designed for both fresh air supply and exhaust air extraction applications. Their compact construction allows for easy floor-mounted installation, while the rock wool insulation integrated between the internal panels provides effective acoustic performance. A dedicated service access panel enables quick and convenient cleaning and maintenance. Combining high efficiency, quiet operation, and ease of maintenance, Turtle cabinet fans are an ideal choice for professional ventilation systems.

Turtle A

Turtle B



Turtle BP



Turtle BPA

Turtle F

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1.7 AIRPORTS

AIRPORTS

Airport terminals are complex structures that require specialized smoke and heat management engineering due to their large atriums, high-ceiling passenger halls, duty-free areas, baggage handling zones, waiting lounges, and high occupant densities. In these facilities, smoke extraction systems are not merely an extension of comfort ventilation; they are critical life-safety systems that directly influence evacuation safety, visibility, toxic gas control, and firefighting operations during a fire emergency.

In North America, these systems are primarily designed in accordance with NFPA 92 – Smoke Control Systems and NFPA 204 – Smoke and Heat Venting. While NFPA 92 focuses on the design, installation, and testing of smoke control systems, NFPA 204 addresses the principles of smoke and heat exhaust ventilation for the emergency removal of combustion products generated during a fire.

In Europe, the primary reference for large-volume facilities such as airports is the EN 12101 Smoke and Heat Control Systems series. In particular, CEN/TR 12101-5 provides design recommendations and calculation methods for atriums, complex buildings, and large-volume spaces. Mechanical smoke exhaust fans are covered by EN 12101-3, smoke barriers by EN 12101-1, smoke control ducts by EN 12101-7, smoke control dampers by EN 12101-8, and power supply systems by EN 12101-10.

Airport Large-Volume Aerodynamics and Smoke Control Principles

The primary objective of smoke control in airport terminals is to maintain the hot smoke layer above occupant level and preserve a safe evacuation height during a fire event. To achieve this, the system creates smoke reservoirs beneath the roof, collects smoke in a controlled manner, extracts it using high-temperature-resistant fans, and maintains pressure balance through controlled make-up air supplied from lower levels. This approach ensures safe evacuation conditions while supporting firefighting operations and maintaining tenable environments for building occupants.

This engineering approach is based on four fundamental principles:

1. Smoke Zoning and Smoke Barriers

To prevent uncontrolled smoke spread throughout the terminal, EN 12101-1 compliant smoke curtains are used to create smoke reservoirs beneath the ceiling. This allows smoke to be contained within designated zones, enabling the extraction system to operate more efficiently and under controlled conditions.

2. Smoke Stratification and Extraction

By utilizing the natural buoyancy of hot smoke, the smoke layer is maintained above occupant level. This layer is then exhausted to the outside through mechanical smoke extraction fans or designated vent openings. In mechanically ventilated systems, smoke exhaust fans must be certified in accordance with EN 12101-3 to ensure high-temperature resistance and sustained performance during fire conditions.

3. Preventing Plugholing (Smoke Layer Penetration)

If smoke extraction is concentrated at a single high-capacity point, the system may draw clean air instead of smoke, leaving the smoke layer trapped within the space. To minimize this risk, multi-point extraction, proper vent placement, suitable exhaust flow rates, and smoke reservoir geometry must be evaluated together. CEN/TR 12101-5 serves as a key design guideline for smoke layer behavior and smoke extraction calculations in large-volume spaces.

4. Make-Up Air Supply

During smoke extraction, excessive negative pressure must be avoided. Otherwise, evacuation doors may become difficult to open, smoke movement may be redirected into unintended areas, and overall system efficiency may decrease. For this reason, controlled fresh air must be supplied from lower levels, with air velocities carefully designed so as not to interfere with occupant evacuation.

NOVVES System Ecosystem

NOVVES provides integrated engineering solutions for smoke and heat exhaust systems in large transportation facilities such as airport terminals, including high-temperature-resistant fans, smoke control dampers, airtight duct connections, and automation-integrated control systems. Upon receiving a signal from the fire detection system, the platform coordinates smoke exhaust fans, dampers, make-up air openings, and associated HVAC equipment to operate as a unified smoke control strategy. In Europe, fire detection and alarm integration is governed by the EN 54 series, while the fire performance classification of smoke control components falls under EN 13501-4.

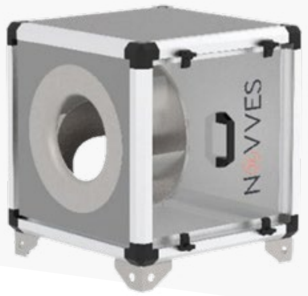
In airport terminals, smoke and heat exhaust systems must be designed through the integrated evaluation of large-volume architectural geometry, smoke layer height, exhaust airflow rates, make-up air balance, fan temperature resistance, damper airtightness, and fire automation controls. By applying the principles of NFPA 92, NFPA 204, and the EN 12101 series, NOVVES delivers highly reliable smoke and heat management solutions that support safe evacuation and effective firefighting operations during fire emergencies.

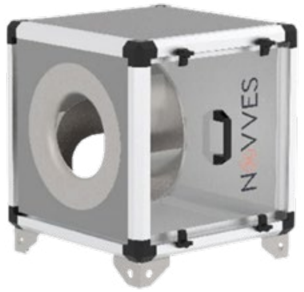
TURTLE
Cabinet Fans

Turtle Series Cabinet Fans are designed for both fresh air supply and exhaust air extraction applications. Their compact construction allows for easy floor-mounted installation, while the rock wool insulation integrated between the internal panels provides effective acoustic performance. A dedicated service access panel enables quick and convenient cleaning and maintenance. Combining high efficiency, quiet operation, and ease of maintenance, Turtle cabinet fans are an ideal choice for professional ventilation systems.


Turtle A


Turtle B


Turtle BP


Turtle BPA


Turtle F

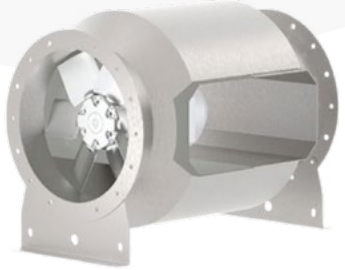
DRAGONFLY
Smoke and Heat Exhaust Fans

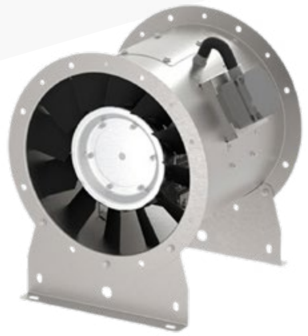
Dragonfly Series is a powerful smoke extraction solution specifically designed for high-temperature fire conditions and certified to F300/F400 fire resistance standards. Engineered to operate continuously for 2 hours at temperatures of 300°C and 400°C, these fans ensure reliable smoke and toxic gas extraction during fire emergencies. In addition to fire applications, Dragonfly fans are widely used in enclosed parking garages and similar environments for the extraction of contaminated air containing carbon monoxide (CO) and hydrocarbons, providing safe and efficient ventilation under demanding conditions.


Dragonfly T


Dragonfly JAU


Dragonfly R


Marlin B


Marlin

MARLIN
Axial Fans

The Marlin Series tubular axial fans are designed for a wide range of ventilation applications, including fresh air supply, exhaust ventilation, and duct-mounted operation. In fire scenarios, they can also function as pressurization fans, preventing smoke from entering escape routes such as stairwells and elevator shafts. Thanks to their compact design, Marlin fans can be installed directly within duct systems, shafts, or rooftop applications. Combining high performance with low noise levels, they provide efficient and comfortable ventilation in a wide variety of environments.

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1.8 WAREHOUSING AND LOGISTICS CENTERS

WAREHOUSING AND LOGISTICS CENTERS

Large logistics centers are among the highest-risk building categories in fire engineering due to their high ceilings, extensive floor areas, and significant fire loads. In these facilities, rapidly spreading toxic smoke poses a greater threat than flames themselves, endangering occupants and potentially leading to structural steel failure. Smoke and Heat Exhaust Ventilation Systems (SHEVS) are therefore critical life-safety components, removing smoke, preserving structural integrity, and facilitating firefighting operations.

Large-Volume Aerodynamics and NFPA 204 Standards

The key life-safety element in large logistics facilities is a mechanical smoke and heat control system designed in accordance with NFPA 204 and NFPA 92 standards.

- **Smoke Stratification Management:** During a fire, hot smoke rises and accumulates beneath the roof, where it is contained within designated smoke zones. The automation system activates exhaust fans and shaft dampers automatically based on signals received from smoke and heat detectors. Fan speeds are controlled through variable frequency drives to prevent plugholing and maintain the integrity of the smoke layer.
- **Management of Large-Volume Spaces:** In lobbies, atriums, and warehouse areas, high-capacity exhaust fans are used to prevent smoke from descending into occupied zones. To avoid plugholing during extraction, smoke is removed through evenly distributed extraction points, while controlled make-up air is supplied to prevent excessive negative pressure and ensure doors remain operable.
- **Technological Reliability:** To ensure flawless system performance, NOVVES provides engineering solutions based on high-reliability, airtight, and safety-focused technologies, supporting a zero-tolerance approach to fire safety.

> **Warehouse ventilation is not designed for comfort alone; it is engineered to preserve visibility, support wayfinding, and provide critical time for emergency response during a fire.**

EN 12101 European Standards and Performance Criteria

Smoke control in logistics centers is a highly engineered process that requires electromechanical equipment to remain operational even under extreme fire conditions. The key elements of this approach include:

Certified Fire Resistance (EN 12101): Smoke exhaust fans must be capable of operating during the most severe stages of a fire (e.g., 400°C for 2 hours) and be certified for high-temperature resistance and continuous performance.

Smoke Curtains and Zoning: To prevent smoke from spreading across vast roof areas and cooling before descending into occupied zones, ceiling-mounted smoke curtains divide the building into dedicated smoke reservoirs. This allows smoke to be contained within specific areas and extracted efficiently by the designated exhaust fans.

Validation Through CFD Simulation: Due to airflow resistance created by racking systems and varying fire loads, system performance must be analyzed using CFD (Computational Fluid Dynamics) simulations. These virtual fire scenarios optimize fan locations and capacities, ensuring effective smoke management under real-world conditions.

Industrial-Grade Reliability: NOVVES provides end-to-end air management solutions supported by a fully certified product ecosystem, delivering 24/7 operational reliability and compliance with international standards for the demanding conditions of logistics facilities.

DRAGONFLY
Smoke and Heat Exhaust Fans

The Dragonfly series consists of F300 and F400 certified smoke and heat exhaust fans designed for operation under extreme fire conditions. Engineered to withstand temperatures of up to 300°C and 400°C for 2 hours, Dragonfly provides reliable performance for both emergency smoke extraction and daily ventilation applications.

Dragonfly T

Dragonfly JAU

Dragonfly R

Dragonfly JAR

Dragonfly JR

Dragonfly C

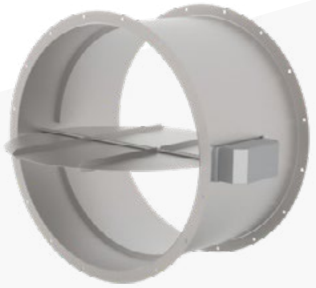
Dragonfly TJF
Tunnel Jet Fans

HOUND
Dampers

Hound Series Dampers are designed to provide maximum control at critical points within ventilation and fire safety systems. Offering a range of solutions including shaft smoke dampers, weather louvers, and pressure relief dampers, the series stands out with its high airtightness, automation compatibility, and durability. By ensuring safe smoke extraction during fire events, directing airflow in outdoor applications, and maintaining pressure balance within ventilation systems, Hound dampers enhance both safety and operational efficiency in modern buildings.

Hound MSD

Hound MSFD-R



HOUND MSFD-B

Koi CB

Koi RB

KOI
Duct Type Fans

KOI Series Duct Fans are designed to meet low- and medium-capacity ventilation requirements. Thanks to their compact construction, they can be easily installed in virtually any position where ventilation is needed. While delivering strong airflow performance, they operate at optimized noise levels to ensure a comfortable indoor environment. Depending on project requirements, KOI fans can be used either as fresh air supply units or as exhaust fans, offering a flexible and efficient ventilation solution.

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1.9

POWER PLANTS & GENERATOR ROOMS

POWER PLANTS & GENERATOR ROOMS

Power plants and transformer substations are strategic infrastructures characterized by high heat loads and elevated fire risks. In these facilities, ventilation systems ensure uninterrupted power generation by preventing critical equipment from overheating, while smoke control systems help protect vital assets during fire incidents. These engineering solutions, which directly determine the operational lifespan and safety of energy facilities, play a crucial role in managing high mechanical and electrical loads.

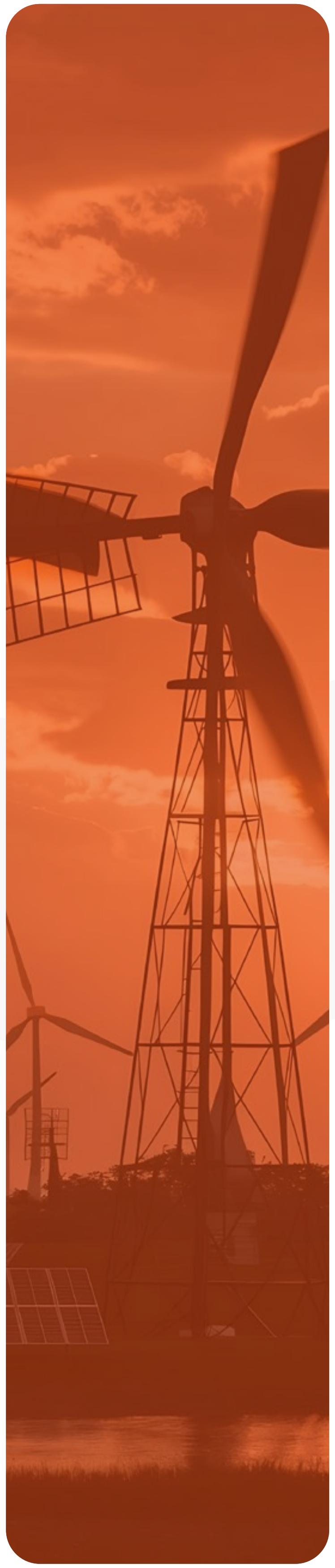
Stack Effect in Boiler and Turbine Halls and Performance-Based Design

- Large vertical spaces within power plants can cause smoke to spread rapidly due to the stack effect. Therefore, performance-based, facility-specific engineering solutions are applied instead of relying solely on standard design rules. Existing ventilation systems are integrated to operate as smoke extraction systems during fire scenarios, while stair towers are protected against smoke infiltration through dedicated pressurization systems.

Continuous Heat and Smoke Extraction in Generator and Transformer Rooms

In enclosed areas with high thermal loads, such as generator rooms and electrical substations, smoke and heat management is based on two fundamental principles:

- **Process Heat Removal:** Continuous mechanical extraction of excessive heat and potential exhaust gases generated by equipment is essential to prevent system failures and insulation degradation.
- **Continuous Operation**: Due to the uninterrupted nature of energy infrastructure, the fans used must be designed for 24/7 operation and equipped with robust heavy-duty industrial motors capable of long-term reliable performance.



CFD Simulations for Design Validation

In high-risk environments such as power plants and generator rooms, smoke and heat control systems are built upon advanced engineering, reliability, and operational resilience.

- **CFD-Based Design Validation:** Complex airflow patterns and thermal distributions are analyzed using Computational Fluid Dynamics (CFD) simulations, often incorporating k-epsilon turbulence models. This approach enables engineers to predict heat loads during both operational and standby conditions, ensuring the ventilation and smoke control system is properly validated before implementation.
- **F300/F400 Certified Fire Resistance:** Due to the high fire loads associated with energy facilities, exhaust fans must comply with EN 12101-3 requirements. These fans must demonstrate continuous operation for 120 minutes at temperatures of 300°C (F300) or 400°C (F400) without performance degradation.
- **Reliability and Operational Continuity:** To support uninterrupted energy infrastructure, industrial ventilation systems are designed for 24/7 operation. This ensures continuous removal of process heat during normal operation while preventing critical system failures and potential blackouts during fire emergencies.

The zero-failure tolerance and heavy-duty operating conditions of power generation facilities cannot be managed with conventional commercial ventilation equipment. NOVVES offers a comprehensive portfolio of industrial ventilation solutions for heat management and emergency smoke extraction in transformer rooms, generator halls, and critical energy infrastructure applications.


Bear W


Bear R


Bear LPF

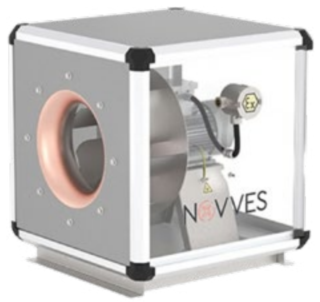
BEAR
Ex-Proof Fans

The Bear Series fans are specifically designed for safe operation in industrial environments where explosive and flammable gases are present, thanks to their Ex-proof motors. Widely preferred in high-risk applications such as chemical plants, wastewater treatment facilities, and paint workshops, the series provides maximum safety through its spark-resistant air inlet cone manufactured from copper material. Combining reliable performance with enhanced explosion protection, Bear Series fans offer a robust ventilation solution for hazardous industrial environments.


Bear EX-C


Bear EX-RRH


Bear EX-RRV


Bear EX-BP

NAUTILUS
Industrial Fans

High-performance industrial fans that combine energy efficiency with powerful airflow, featuring backward-curved impellers and direct-drive motor technology for reliable and efficient ventilation. They are ideally suited for extracting air containing dust, wood chips, oil mist, and similar particulates, while also serving as an effective solution for projects requiring fresh air supply. Known for their quiet operation, these fans offer flexible airflow management thanks to their adjustable discharge direction, allowing airflow to be oriented as needed. Durable, versatile, and engineered for demanding industrial environments, they provide a dependable ventilation solution for a wide range of applications.


Nautilus Silence


Nautilus LFP

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HOTELS AND RESIDENTIAL HIGH-RISE BUILDINGS

1.10

HOTELS AND RESIDENTIAL HIGH-RISE BUILDINGS

Modern high-rise buildings, iconic symbols of metropolitan skylines, rely on a phased evacuation strategy during fire emergencies, as evacuating the entire building simultaneously is often impractical. Instead, only the affected floors are evacuated. The success of this strategy depends on keeping escape stairwells and elevator shafts completely smoke-free through Pressure Differential Systems (PDS). These systems create protected vertical evacuation routes, allowing occupants in unaffected areas to remain safely in place while preventing smoke from spreading throughout the building.

High-Rise Building Aerodynamics and the "Stack Effect"

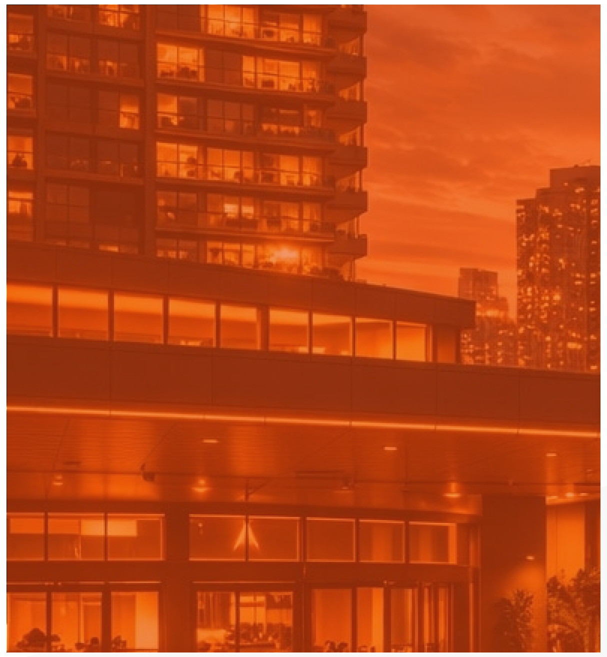
One of the most significant thermodynamic challenges in high-rise fire safety is the Stack Effect, caused by temperature differences between the inside and outside of the building. In this phenomenon, heated air rapidly rises through vertical shafts, carrying smoke to upper floors within seconds and potentially blocking evacuation routes. To counteract this risk, Pressurization Systems (PDS) supply clean air to escape routes, creating a positive pressure environment that physically prevents smoke from entering these protected areas.

Pressure Differential Criterion: A sufficient pressure difference must be maintained between the escape stairwell and the fire floor to prevent smoke infiltration. This pressure differential is typically designed within a range of 30 Pa to 50 Pa.

Door Opening Force Criterion: In stairwell pressurization systems, the supplied air pressure must not only prevent smoke leakage but also allow evacuation doors to be opened safely. Excessive pressure can make doors difficult to open, particularly for children, elderly occupants, and individuals with limited mobility.According to the American approach under NFPA 92 and NFPA 101, the required door opening force should generally not exceed 30 lbf (approximately 133 N). Under the European approach, EN 12101-13:2022 specifies that the opening force measured at the door handle along an escape route must not exceed 100 N while the pressurization system is operating.To maintain these limits, pressurization systems should be controlled using pressure sensors, relief dampers, variable-frequency drive (VFD) fans, and automatic balancing algorithms.

Corridor Smoke Shafts and Smoke Extraction

In luxury hotels and residential towers, smoke extraction from windowless corridors is achieved through mechanical smoke shafts and high-capacity roof-mounted exhaust fans. During a fire event, only the smoke damper serving the affected floor is opened, allowing toxic gases to be extracted from the building through negative pressure. Keeping dampers on all other floors closed creates a critical safety barrier that prevents smoke migration between levels.



For many years, pressurization and smoke control systems in high-rise buildings throughout Europe were designed in accordance with EN 12101-6. Today, this standard is being superseded by EN 12101-13 (Pressure Differential Systems), which establishes significantly more rigorous requirements for system design, installation, commissioning, testing, and ongoing performance verification. The updated standard requires systems to perform automatic self-diagnostics, continuously monitor operational status, and immediately report faults, enhancing life safety performance in high-rise buildings to the highest level.

In hotel and residential developments, smoke control systems must deliver more than fire protection alone. In addition to ensuring safe evacuation, they must also provide acoustic comfort by maintaining low noise levels within shafts, service areas, and suspended ceiling spaces. NOVVES offers a specialized portfolio of solutions engineered to meet the sophisticated architectural demands and stringent safety requirements of modern high-rise buildings.

"In high-rise buildings, ventilation is not designed for comfort alone; it is engineered to ensure that occupants can evacuate the building safely during an emergency.

DRAGONFLY

Smoke and Heat Exhaust Fans

The Dragonfly series consists of F300 and F400 certified smoke and heat exhaust fans designed for operation under extreme fire conditions. Engineered to withstand temperatures of up to 300°C and 400°C for 2 hours, Dragonfly provides reliable performance for both emergency smoke extraction and daily ventilation applications.

Dragonfly T

Dragonfly JAU

Dragonfly R

Dragonfly JAR

Dragonfly JR

Dragonfly C

Dragonfly TJF
Tunnel Jet Fans

HOUND

Dampers

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Hound MSD

Hound MSFD-R

Hound MSFD-B

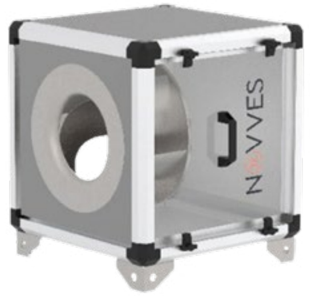
The Turtle Series cabinet fans are designed for both fresh air supply and exhaust air extraction applications. Their compact construction allows for easy floor-mounted installation, while the rock wool insulation integrated between the inner panels provides effective acoustic attenuation. The integrated access panel enables quick and convenient cleaning and maintenance operations. Combining high efficiency, quiet operation, and user-friendly maintenance, Turtle Series cabinet fans are an ideal choice for professional ventilation systems requiring reliable performance and acoustic comfort.

TURTLE

Cabinet Fans

Turtle A

Turtle B



Turtle BP

NOVVES

BORN TO FLOW.

Shaping the Invisible

"If a city breathes, it is because sound engineering is at work."

ŞİRKET PROFİLİ TR
COMPANY PROFILE TR

ŞİRKET PROFİLİ EN
COMPANY PROFILE EN

ŞİRKET PROFİLİ RU
COMPANY PROFILE RU

REFERANS TR-EN
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SERTİFİKALAR RU
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BIOMIMETICS TR-EN

BİYOMİMETİK RU
BIOMIMETICS RU

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SOLUTIONS TR-EN

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